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Progressive Lead-Time Forecasting of Agricultural Yields from Weather Data:

Evidence from County-Level U.S. Soybean Yields

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1. Introduction

1.1 Problem, Practical Value, and Motivation

Agricultural yield becomes observable at harvest, after many operational and commercial commitments have been executed. Earlier forecasts provide essential decision support for risk management in agrifood supply chains. The operational value of an agricultural forecast depends directly on its lead time: pre-sowing projections support acreage allocation, input procurement, and long-term storage planning, whereas in-season updates inform regional sourcing adjustments, freight commitments, and hedging strategies. We evaluate predictive skill across three criteria: accuracy across the full yield distribution, the operational timing at which genuine predictive signal emerges, and the stability of predictive gains across successive monthly updates from the pre-season origin to harvest eve ($H = 12 \dots 1$).

Within global crop markets, soybean (*Glycine max*) represents a vital commodity underpinning domestic livestock feed rations and international trade flows, with the Midwestern Corn Belt serving as the primary production engine (Schlenker and Roberts, 2009). Harvest surprises in this region carry severe, asymmetric economic costs. Regional crop shortfalls, typically driven by summer heatwaves or moisture deficits, generate acute procurement bottlenecks and sharp price spikes for grain crushers, feed millers, and export terminals. Empirically, across the 1960–2025 record, each 10% negative departure from secular yield trend is associated with an average 7.5% increase in annual commodity prices ($p = 0.014$), with intra-seasonal futures rallies on the Chicago Board of Trade (CBOT) frequently exceeding 30–60% during acute summer droughts. Conversely, unexpectedly bumper harvests compress local storage capacity, create barge and rail bottlenecks along the Mississippi corridor, and exert downward pressure on regional cash basis levels.

We focus specifically on county-level yield anomalies (in bushels per acre), which isolate climate sensitivity from secular technology and cross-county acreage shifts. Aggregate production totals depend on land allocation decisions, while commercial prices reflect global carryover stocks and international trade flows. These supply-chain repercussions establish the practical motivation for county-level yield anomaly forecasting.

1.2 Research Question and Target Formulations

A fundamental empirical challenge in agricultural forecasting is that raw county yields are heavily dominated by secular technological progress, including genetic breeding, synthetic fertilizers, chemical pest control, and farm mechanization. To isolate how much and when observed surface weather explains realized crop output, we must first decouple interannual

weather sensitivity from the underlying secular technological trajectory.

The central research question asks: *At what lead times prior to harvest does observed surface weather add statistically significant and stable out-of-sample predictive skill for county-level soybean yield anomalies, and how does this skill vary across model families and temporal aggregations?*

To address this question, we structure our predictive framework around two complementary evaluation dimensions:

- **Continuous Anomaly Forecasting (Regression):** We estimate continuous models predicting county-level yield anomalies, isolating weather-induced departures from secular technological trends fitted strictly on historical training folds. Absolute yield levels are subsequently reconstructed by adding predicted anomalies to the estimated secular trend.

- **Downside Tail-Risk Stress Testing:** In agricultural supply chains, forecast errors are economically asymmetric: standard quadratic loss near historical averages is manageable, whereas failing to anticipate extreme adverse shortfalls causes severe contractual defaults and market disruptions. We stress-test continuous regression models specifically on extreme negative anomalies (the bottom decile D_1) to assess whether empirical estimators capture severe climate shocks or suffer from excessive shrinkage toward historical means.

We structure our investigation around five core research questions and formal hypotheses (H1–H5, derived in Chapter 2): evaluating incremental weather skill over secular trends, mapping lead-time persistence across horizons ($H = 12 \dots 1$), assessing sub-monthly temporal aggregation, comparing linear parsimony against non-linear tree and deep learning architectures, and stress-testing the anticipation of downside tail risks.

1.3 Methodological Contributions and Scope Boundaries

The literature on empirical crop modeling encompasses diverse approaches to predicting county-level soybean yields from environmental data. Recent studies have advanced the field from complementary perspectives: Vijverberg et al. (2023) identified pre-sowing climate precursors for extreme yield events; Yin et al. (2026) demonstrated the influence of sub-monthly temporal aggregation on modern architectures over recent decades; and the multidecadal benchmark by Chen and Zhang (2026) examined in-season deep learning across monthly growing-season updates.

However, existing empirical studies examine these dimensions in isolation, focusing on specific phenological windows, individual model classes, or restricted historical intervals. We synthesize these empirical branches into a unified benchmarking framework.

We address these gaps through two primary methodological contributions:

1. **Systematic 12-Month Lead-Time Mapping Without Forecast Inputs ($H = 12 \dots 1$):** We establish the empirical boundaries of weather-driven predictability across twelve standardized monthly cutoffs from the preceding autumn through harvest eve, tracking precisely when observed atmospheric states transition from dormant noise into reliable predictive skill. To guarantee causal realism and avoid look-ahead bias, our operational setup relies strictly on cumulative reanalysis observations available up to each forecast origin without numerical weather predictions, eliminating forecast drift and calibration artifacts while isolating the empirical contribution of observed weather.

2. **Comparison of Forecasting Specifications:** We compare regularized linear models, tree ensembles (Random Forest, XGBoost), and recurrent architectures (LSTM) on the selected county panel. The final benchmark evaluates monthly tabular models and a 10-day LSTM. The recurrent frequency and head-specific parameters are fixed before the final test evaluation.

We evaluate this framework on soybean across a balanced panel of 135 counties in six core Midwestern states (Illinois, Indiana, Iowa, Minnesota, Missouri, and Ohio) over 1951–2025 (10,125 county-year observations). This geographic and temporal delimitation provides an unbroken historical record that avoids artificial data imputation.

We interpret feature importance strictly as empirical predictive association, without claiming structural causal identification.

Weather-only models provide the initial benchmark. We then test the incremental contribution of lagged yield anomalies and retain integrated specifications supported by that comparison. Reported scores distinguish the two information sets; lagged anomalies are detrended on training data and included only when available at the forecast origin.

1.4 Structure of the Study

Chapter 2 reviews the literature and derives H1–H5. Chapter 3 describes the data and exploratory analysis. Chapter 4 presents the methodology. Chapter 5 reports the empirical results, and Chapter 6 synthesizes findings, evaluates hypotheses, and concludes the study. Appendix A provides sensitivity analyses on structural breaks and detrending specifications.

2. Literature Review

2.1 Technological Trends, Weather Variability, and Soybean Agronomic Sensitivity

In agricultural economics, crop yield (Y_{it}, bushels per acre) serves as the primary metric of biological land productivity. Across multidecadal horizons, developed agricultural systems exhibit strong upward trajectories: in the U.S. Corn Belt, average soybean yields more than doubled from approximately 20 bushels per acre in the early 1950s to over 50 bushels per acre today, driven by cumulative genetic improvements, chemical inputs, mechanization, and precision farming. Consequently, historical yield series Y_{it} decompose into two additive components:

$$Y_{it} = T_{it} + \varepsilon_{it} \tag{2.1}$$

where T_{it} represents the smooth technological trend capturing non-meteorological advancements, and ε_{it} denotes the interannual yield anomaly driven by within-season weather variability, transient pest pressures, and environmental shocks.

Failing to account for T_{it} introduces severe statistical pathologies. Because atmospheric greenhouse gas concentrations and temperatures have trended upward over the same multidecadal period, estimating models directly on raw yields risks confounding biological weather sensitivity with secular technological growth (Schlenker and Roberts, 2009). A model estimated on raw yields may spuriously attribute genetic gains to rising temperatures, distorting both predictive accuracy and feature attributions. Isolating the weather-driven anomaly ε_{it} is therefore essential for characterizing meteorological impacts.

2.1.1 Non-Linear Agro-Climatic Stress Indicators

Soybean is an annual legume whose physiological development and final seed production respond non-linearly to prevailing meteorological conditions, governed by acute extremes and phenological exposure timing. Within moderate ranges, temperature accelerates vegetative growth, leaf expansion, and developmental progression. These cumulative thermal requirements are represented via Growing Degree Days (GDD), accumulated between a base temperature (10°C) and an upper threshold (30°C). Hoffman et al. (2020) demonstrate that tracking thermal time provides a biologically grounded framework for identifying developmental milestones, capturing yield-limiting stress far more effectively than standard calendar time.

When temperatures exceed critical thresholds, thermal damage accelerates rapidly. In a seminal empirical study, Schlenker and Roberts (2009) demonstrated sharp nonlinear breakpoints: while temperatures up to 29°C or 30°C are beneficial, temperatures

exceeding this threshold inflict severe yield penalties via reproductive heat stress, floral abortion, and accelerated canopy senescence. To capture these damage mechanisms, researchers construct Extreme Degree Days (EDD), accumulating thermal exposure strictly above physiological damage thresholds ($> 30°C$). Coarse monthly temperature averages smooth out these excursions, masking heatwaves lasting only a few consecutive days. Both Hoffman et al. (2020) and Chen and Zhang (2026) identify indicators of extreme summer heat as among the most decisive predictors of interannual yield variance.

While rainfall replenishes soil moisture, crop water stress is governed by the joint interaction between soil water availability and atmospheric evaporative demand, characterized by Vapor Pressure Deficit (VPD). Hoffman et al. (2020) identify elevated VPD as a critical driver of yield reduction during reproductive growth: high VPD forces stomatal closure to prevent desiccation, halting transpirational cooling and carbon fixation even when soil moisture is present. Regarding the surface water budget, empirical studies emphasize compounding interactions between precipitation, root-zone soil moisture, and atmospheric demand (Chen and Zhang, 2026; Hoffman et al., 2020; Vijverberg et al., 2023). We parameterize the surface climatic water balance via the difference between accumulated precipitation and reference evapotranspiration ($P - \mathrm{ET}_0$), calculated following standard FAO-56 Penman-Monteith formulations. Because biological damage concentrates during midsummer reproductive flowering and pod filling, detailed agronomic descriptions of these phenological vulnerability windows are formalized in Section 3.1.2.

2.2 Empirical Design Dimensions: Lead Times and Baseline Benchmarking

2.2.1 Forecasting Horizons and In-Season Updating

In traditional agricultural impact modeling, studies predominantly evaluate crop responses using post-harvest weather series aggregated across the complete growing season (Schlenker and Roberts, 2009). While indispensable for retrospective structural estimation, such post-harvest models provide limited advance notice for operational decision-makers in agricultural supply chains.

Recent forecasting literature has recognized that predictive utility depends directly on forecast lead time, shifting toward progressive in-season updating. For instance, Chen and Zhang (2026) evaluate monthly models from May through August, demonstrating that explained variance increases progressively as reproductive weather unfolds. In the U.S. Corn Belt, the critical reproductive phases of flowering (R1–R2) and seed filling (R5–R6) conclude biologically by late August; studies focused strictly on agronomic response mechanisms terminate observations at that point because biological yield accumulation is largely complete. Conversely, operational supply-chain applications require extending predictions through harvest eve in October ($H = 1$, or September $H = 2$ for early-

maturing northern states). This 12-month continuum covers the full commercial decision window and allows formal testing of whether late-season conditions contribute actionable skill or introduce post-maturity noise. At earlier horizons, Vijverberg et al. (2023) demonstrate that antecedent land and ocean states provide predictive signals up to 50 days prior to planting. Evaluating models across sequential horizons ($H = 12 \ldots 1$) maps precisely when atmospheric observations transition from uninformative noise into reliable forecasting signals.

2.2.2 Reference Baselines and Benchmark Anchoring

In empirical forecasting, evaluating a model's true capability requires benchmarking against parsimonious reference baselines: historical county climatology, deterministic linear trends, or lagged past yields ($Y - 1$).

In trending agricultural panels, baseline models capture the substantial variance attributable to secular technological momentum and permanent cross-sectional productivity differences across counties. Incremental weather skill is assessed through out-of-sample error reductions relative to these baselines. These reductions establish predictive contribution within the evaluation sample; operational value requires a separate assessment. For example, in ablation experiments by Chen and Zhang (2026), incorporating lagged yield ($Y - 1$) accounted for approximately half of the model's total explained variance, highlighting the essential role of baseline decoupling when isolating pure weather sensitivity.

2.3 Statistical, Machine Learning, and Sequential Model Architectures

The literature features three prominent model families for tabular and sequential crop forecasting:

Regularized Linear Models. Ridge and Lasso regressions provide transparent, computationally efficient baselines that mitigate multicollinearity among highly correlated meteorological features, serving as robust benchmarks against which more complex nonlinear architectures can be judged (James et al., 2023).

Tree-Based Ensembles. Random Forests and Gradient Boosted Decision Trees (XGBoost) represent predominant workhorses for tabular agro-meteorology (Sharma et al., 2025). They naturally accommodate step-function damage thresholds (such as non-linear heat stress above 30°C) and multi-way feature interactions without restrictive parametric assumptions, while facilitating post-hoc feature attribution via TreeSHAP (Chen and Zhang, 2026).

Sequential Deep Learning. Recurrent architectures, particularly Long Short-Term Memory (LSTM) networks, are designed to process time-series sequences of meteorological inputs, dynamically capturing stress duration, dry spell persistence, and intra-seasonal shock timing (Khaki et al., 2020).

Temporal Granularity in Feature Representation. Beyond model architecture, the temporal granularity of meteorological features represents a critical design choice. While traditional benchmarks rely on monthly averages (Chen and Zhang, 2026), sub-monthly aggregations (such as 5-day and 10-day windows) preserve acute, short-duration weather extremes that monthly means smooth out (Yin et al., 2026).

2.4 Approaches to Unobserved Future Weather at Forecast Origin

Forecasting studies handle unobserved weather after the cutoff through three approaches:

1. **Observed-weather truncation.** Horizon-specific models use observations available through the cutoff. Chen and Zhang (2026) apply monthly cutoffs; related weekly and sub-monthly designs are reviewed below.

2. **Dynamical seasonal forecasts.** Numerical forecasts supply the remaining weather sequence. Ceglar and Toreti (2021) use ECMWF SEAS5; crop predictions inherit seasonal forecast uncertainty and spatial biases.

3. **Climatological or synthetic completion.** Historical averages or weather generators complete future weeks, as in process-based crop modelling. Completion assumptions influence the resulting yield predictions.

We use observed-weather truncation to assess how predictive skill changes as weather information accumulates. This choice excludes seasonal forecasts and synthetic completion from the empirical comparison.

2.5 Benchmark Studies and Comparative Analysis

To position our empirical framework, we examine primary benchmarks in multi-lead-time county crop forecasting across the United States Corn Belt.

2.5.1 Chen & Zhang (2026): Multidecadal County-Level Deep Learning Benchmark

Chen and Zhang (2026) compile a 99-year panel (1927–2025) across 1,309 U.S. counties, combining USDA NASS soybean yields with 31 monthly weather variables and static SSURGO soil features. They train a Deep Neural Network and a Random Forest, evaluating monthly forecasts from May through August. On holdout test campaigns (2024–2025),

end-of-season R^2 reaches 0.67–0.68, with August extreme heat days ($> 32°C$) driving yield reductions. Omitting lagged yield ($Y - 1$) contracts test R^2 to ≈ 0.33, showing that historical baseline persistence accounts for half of the explained variance in raw yield levels.

However, their empirical design exhibits structural limitations: the apparent accuracy of un-detrended raw yield models largely absorbs secular technology trends, obscuring atmospheric sensitivity, while evaluating holdouts on only two recent test years leaves the temporal stability and generalization of deep learning architectures across diverse climatic regimes unverified.

2.5.2 Vijverberg et al. (2023): Presowing Precursors and Extreme Event Classification

Vijverberg et al. (2023) investigate presowing predictability up to 50 days before planting. Using logistic regression and Random Forests driven by Pacific sea surface temperatures and antecedent soil moisture, they predict poor-yield events (anomalies below the 33rd percentile), achieving regional classification skill (ROC-AUC ≈ 0.65–0.75).

While establishing the value of antecedent climatic states, framing the task as a binary classification problem discards continuous shock magnitudes, preventing direct estimation of commercial supply volumes and quantitative downside risks.

2.5.3 Yin et al. (2026): Temporal Aggregation and Sequential Architectures

Yin et al. (2026) evaluate temporal aggregation granularity (5-day, 10-day, 15-day, and 30-day summaries) across Midwest counties (2000–2020) using CNN, LSTM, and Random Forest architectures. They demonstrate that sub-monthly blocks (5-day and 10-day) preserve acute thermal and hydrologic shocks during flowering and pod filling that monthly means smooth out.

Nevertheless, their panel is restricted to a short 21-year satellite-era window, limiting historical exposure to rare extreme macro-climatic disasters and leaving open how high-frequency representations perform under multidecadal expanding-window validation.

2.5.4 Khaki et al. (2020) and Complementary Frameworks

Khaki et al. (2020) deploy hybrid CNN-RNN models with weekly in-season updating for Midwest corn and soybeans (1980–2018), documenting steady error reductions toward harvest. Complementing this, Iizumi et al. (2021) show that within-season shortfall probabilities can be anticipated several months ahead.

These benchmarks assume the architectural superiority of complex deep networks a priori, without conducting formal parsimony tests or evaluating whether regularized linear models with agronomic domain features could achieve comparable accuracy under strict chronological validation.

2.6 Synthesis, Literature Gap, and Hypotheses

2.6.1 Literature Consensus and the Identified Opportunity

The reviewed literature establishes strong consensus on non-linear crop weather sensitivity, the primacy of midsummer reproductive vulnerability, and the utility of in-season updating. However, existing studies leave four decisive methodological gaps:

1. **Decoupling Anomaly from Secular Trend:** Pervasive reliance on un-detrended raw yields conflates technological momentum with atmospheric sensitivity, inflating apparent model skill via autoregressive lags.

2. **Unified 12-Month Predictive Continuum:** Empirical efforts remain fragmented between pre-sowing climate precursors and isolated in-season windows, lacking a continuous monthly mapping from overwinter dormancy through harvest eve without look-ahead bias or numerical forecast drift.

3. **Multidecadal Sub-Monthly Sequence Validation:** High-frequency temporal representations (10-day decads) have been tested only over short recent periods, leaving their empirical stability unverified over multidecadal historical panels.

4. **Rigorous Parsimony and Tail-Risk Validation:** Deep learning architectures are widely assumed superior despite lacking formal out-of-sample significance tests (e.g., Diebold–Mariano) against penalized linear baselines, while their capacity to anticipate catastrophic downside tail risks remains unverified.

2.6.2 Derivation of Research Hypotheses

To resolve these four gaps, we structure our empirical inquiry around five core research questions and formal hypotheses:

- **Hypothesis 1 (Incremental Weather Skill):** *Does observed surface weather add genuine predictive value over historical climatology and secular trends?* We hypothesize that incorporating high-resolution ERA5-Land reanalysis produces a statistically significant reduction in out-of-sample forecast error for county yield anomalies relative to deterministic trend baselines.

- **Hypothesis 2 (Lead-Time Dependence and Persistence):** *At what lead time before harvest does reliable predictive skill emerge, and does it persist across subsequent updates?* We hypothesize that predictive skill emerges during midsummer reproductive growth and increases monotonically as harvest approaches ($H \to 1$), with early gains persisting without reverting to noise.

- **Hypothesis 3 (LSTM Temporal Aggregation):** *Does a 10-day sub-monthly temporal resolution provide measurable forecasting value compared to standard 30-day monthly summaries across a multidecadal panel?* We hypothesize that 10-day decad sequences capture short-duration heat and moisture extremes, significantly improving out-of-sample validation accuracy over monthly summaries when recurrent architecture hyperparameters are held fixed.

- **Hypothesis 4 (Model Complexity under Strict Validation):** *Do complex sequential deep learning networks systematically outperform parsimonious regularized linear models under strict chronological validation?* We hypothesize that under strict expanding-window validation on interannual anomalies, highly parameterized deep learning models do not systematically outperform transparent regularized linear models (ElasticNet) or tree ensembles.

- **Hypothesis 5 (Tail-Risk and Downside Shock Anticipation):** *Can continuous regression models anticipate severe agricultural disasters (the bottom decile D_1), or do they collapse toward historical conditional means?* We hypothesize that continuous empirical regression models successfully anticipate acute downside yield shocks prior to harvest, correctly resolving directional shortfall regimes while mitigating conditional shrinkage.

3. Data and Study Area

The empirical analysis rests on two primary data streams spanning the United States Midwest from 1951 to 2025: county-level soybean yields from the USDA National Agricultural Statistics Service (NASS), and high-resolution gridded surface weather from the ECMWF ERA5-Land reanalysis.

3.1 Agronomic Setting and County Panel Assembly

3.1.1 County Panel Selection Rationale and Geographic Scope

We focus on six core states of the United States Corn Belt: Illinois, Indiana, Iowa, Minnesota, Missouri, and Ohio. These states represent the primary production center for domestic soybeans, accounting for the highest harvest volumes and acreage density in the country. Crop production across this domain is predominantly rainfed, with commercial irrigation covering less than 5% of harvested area (Schlenker and Roberts, 2009). Realized yields directly reflect natural weather variability, temperature extremes, and seasonal moisture deficits.

We compile county-level soybean yields (in bushels per acre, 1 bu/ac $\approx$ 67.25 kg/ha) from the USDA NASS Quick Stats database. To assemble the balanced panel, we require (i) a complete annual record for 1951–2025 without imputation and (ii) high harvested density.

We prioritize counties with high harvested density because ERA5-Land weather is aggregated over county polygons. This selection increases the agricultural relevance of county averages, but the polygon weights do not isolate cropland microclimates.

While ERA5-Land begins in January 1950, constructing the complete 12-month campaign for crop year 1950 would require reanalysis records from November 1949. Consequently, 1951 is the first crop year with complete antecedent weather data.

This protocol isolates a balanced panel of **135 counties** over 75 years ($135 \times 75 =$ **10,125** observations). Table 3.1 and Figure 3.1 detail their spatial coverage across the primary agro-climatic subregions of the Corn Belt.

3.1.2 Soybean Phenology and Critical Agro-Climatic Sensitivity Windows

Soybean physiological development and yield response exhibit strong temporal heterogeneity across the growing cycle. Meteorological drivers vary fundamentally in sign and magnitude depending on phenological timing: while early vegetative stages tolerate environmental fluctuations through root expansion and canopy plasticity, the summer reproductive window (July–August) constitutes the primary biophysical bottleneck. Heat extremes ($T_{\max} > 30$°C) and elevated vapor pressure deficit ($VPD > 1.5$ kPa) during

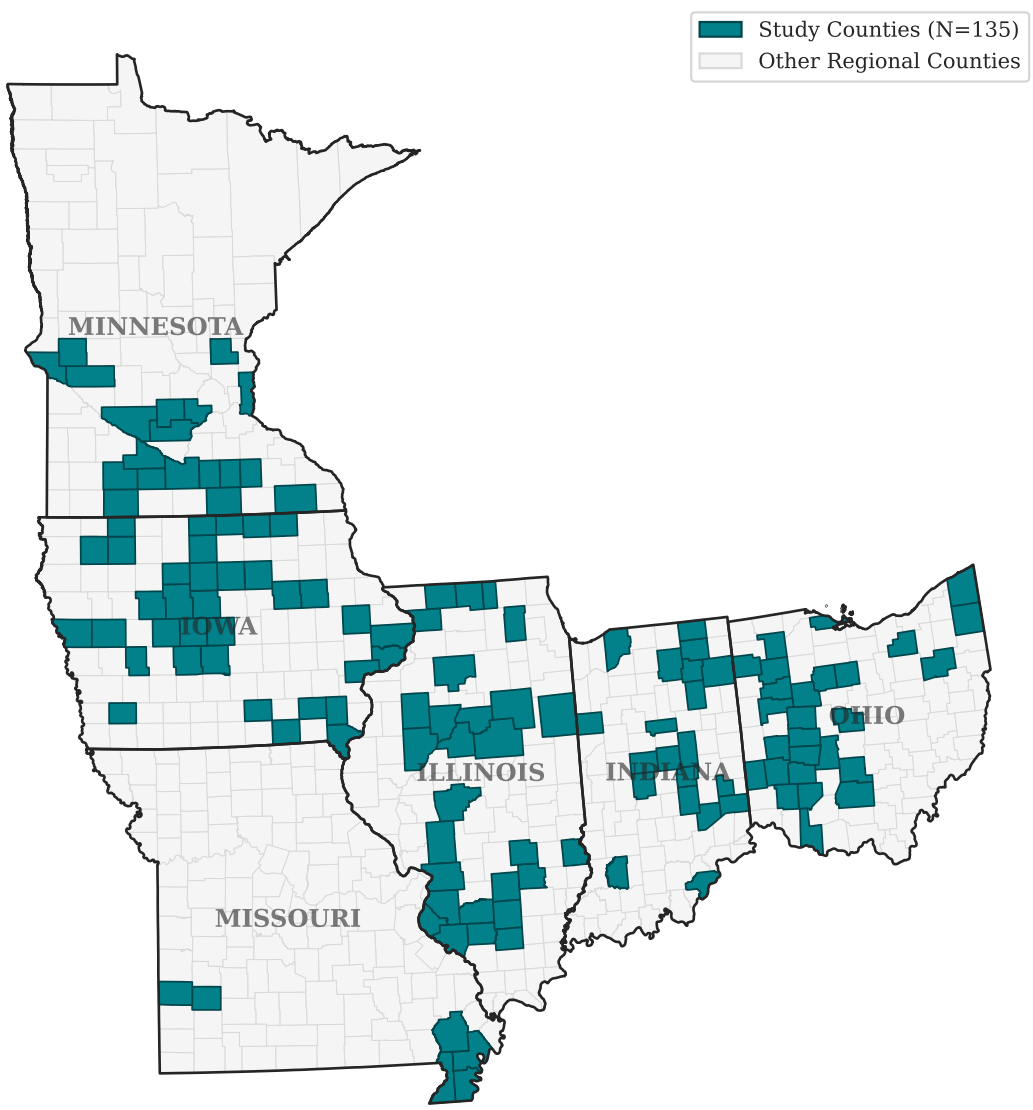


Figure 3.1. Geographic distribution of the 135 study counties across the six Midwestern states (1951–2025), illustrating spatial coverage across the primary agro-ecological subregions of the Corn Belt.

Source: Author's calculation on USDA NASS and US Census TIGER/Line data.

Table 3.1. Descriptive summary of the 135 study counties across the six Midwestern states (1951–2025).

State	Counties (N)	Observations ($N \times 75$)	Mean Yield (bu/ac)	Std. Dev. (bu/ac)
Illinois	28	2,100	38.35	13.20
Indiana	20	1,500	39.52	11.82
Iowa	35	2,625	39.73	12.01
Minnesota	19	1,425	35.10	12.70
Missouri	6	450	29.04	10.54
Ohio	27	2,025	36.88	11.40
Panel Total	**135**	**10,125**	**37.72**	**12.40**

Notes: Counties (N) indicates the number of counties with complete annual records from 1951 to 2025 without imputation. Mean Yield and Std. Dev. are in bushels per acre. *Source:* Author's calculation on USDA NASS Quick Stats data (1951–2025).

flowering trigger floral abortion and pod drop, whereas moisture deficits during seed filling permanently constrain seed weight. This concentrated sensitivity motivates evaluating forecasting skill across twelve discrete monthly horizons and testing high-frequency 10-day decad representations.

Regional maturity gradients further govern harvest timing across the Corn Belt. Cultivar adoption follows a strict latitudinal progression of Maturity Groups (Hoffman et al., 2020): northern Minnesota relies on early-maturing Group I varieties that reach physiological maturity (R7) by mid-September, with harvesting largely concluded before October. Across central and southern states (Groups II–IV), maturation and combining extend through late October. This biological divergence provides the agronomic rationale for implementing a state-specific harvest closure rule ($H^*(s) = 2$, September 30 for Minnesota vs $H^*(s) = 1$, October 31 for the remaining five states), eliminating uninformative post-harvest weather noise.

3.2 Exploratory Data Analysis of County Soybean Yields (1951–2025)

3.2.1 Secular Technological Trend, Productivity Tripling, and Anomaly Decomposition

To isolate climatic shocks from secular agronomic gains, we formally decompose realized county soybean yield $y_{c,t}$ (bu/ac for county c and year $t \in \{1951, \ldots, 2025\}$) into two additive components:

$$y_{c,t} = \tau_{c,t} + \epsilon_{c,t} \tag{3.1}$$

where $\tau_{c,t}$ represents the deterministic secular trend driven by technological, agronomic, and genetic progress, and $\epsilon_{c,t}$ denotes the stochastic yield anomaly induced by weather

variability and environmental stress.

Our machine learning models in Chapter 4 are trained exclusively to predict the detrended yield anomaly $\epsilon_{c,t}$ (or the detrended baseline $\hat{y}_{c,t}^{\text{detrended}} = \bar{y}_c + \epsilon_{c,t}$), removing the predictable secular trend. In operational validation, we estimate trend parameters strictly on historical training folds to prevent look-ahead bias; here we examine full-sample properties to evaluate trend specifications.

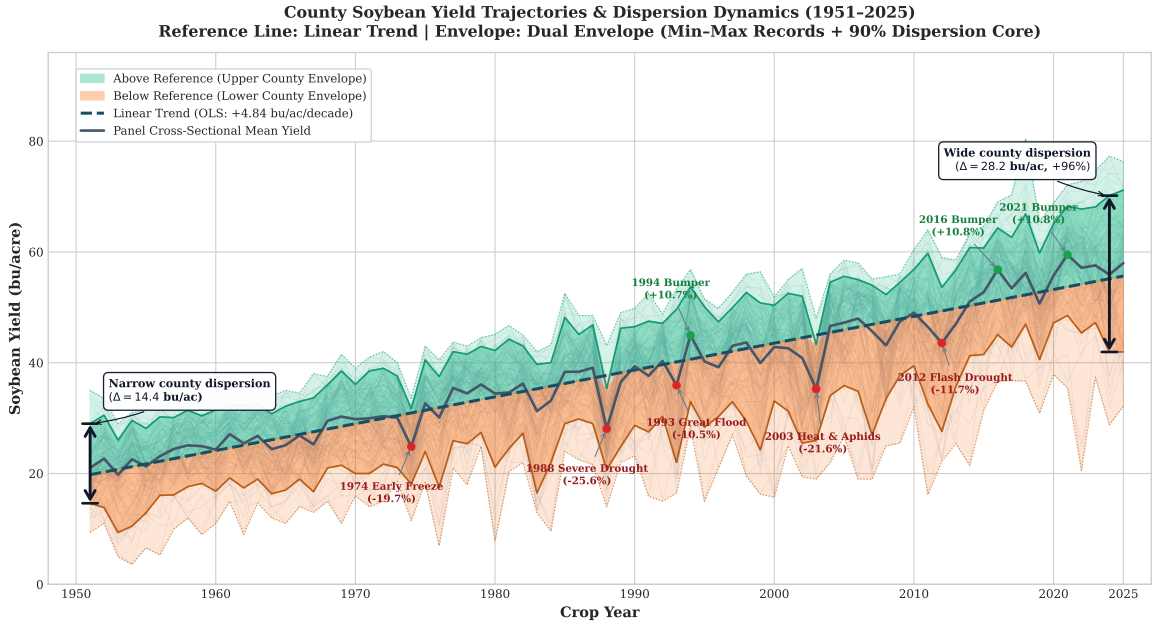


Figure 3.2. County soybean yield trajectories, secular linear trend, and cross-county dispersion envelopes across 135 Midwestern counties (1951–2025). Shaded envelopes delineate the inner 90% cross-county dispersion core and historical min–max record boundaries relative to the long-term trend; the solid line traces the annual cross-sectional panel mean.

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data.

Regional mean yields more than doubled over the panel (Table 3.2). At the regional annual level, an ordinary least squares (OLS) regression estimates an average gain of 0.484 bu/ac/year and explains 93.23% of variation (Figure 3.2):

$$\hat{y}_t = 0.484 \cdot t - 925.46 \quad (t = 1951, \ldots, 2025) \tag{3.2}$$

The regional fit combines technological drift with spatial averaging: local departures partly offset across counties. County-specific trends account for heterogeneous productivity growth, including the state differences described in Section 3.2.2. Across the full county panel, pooled explained variance is lower ($R^2 = 0.8122$). The separate expanding-window comparison below evaluates the predictive adequacy of county-level trend specifications.

Absolute yield dispersion also increases over time, despite a declining coefficient of variation. The annual cross-county spread expands by over 70%, and the 90% dispersion envelope widens by more than 50% (Table 3.2). Thus, a similar proportional shortfall

entails a larger physical loss at modern yield levels. These properties motivate county anomaly forecasting and the subsequent examination of weather sensitivity.

Table 3.2. Decadal distributional dynamics of Midwestern county-level soybean yields (1951–2025).

Period	Obs. (N)	Mean (bu/ac)	Median (bu/ac)	Std. Dev. (bu/ac)	Min (bu/ac)	Max (bu/ac)	IQR (bu/ac)	CV (%)
1950s (1951–1959)	1,215	22.77	23.10	5.01	3.6	35.0	6.55	22.0%
1960s (1960–1969)	1,350	26.51	26.60	5.07	9.0	41.5	6.98	19.1%
1970s (1970–1979)	1,350	31.38	32.00	6.04	7.0	44.5	8.90	19.2%
1980s (1980–1989)	1,350	35.01	35.75	6.63	7.9	52.5	8.58	19.0%
1990s (1990–1999)	1,350	40.43	41.50	7.08	15.0	56.8	9.27	17.5%
2000s (2000–2009)	1,350	43.97	45.00	7.78	15.7	58.5	11.20	17.7%
2010s (2010–2019)	1,350	50.70	51.40	7.80	16.2	80.4	10.00	15.4%
2020s (2020–2025)	810	57.31	57.95	7.57	20.6	77.3	9.65	13.2%
Full Panel (1951–2025)	**10,125**	**37.72**	**36.70**	**12.40**	**3.6**	**80.4**	**19.00**	**32.9%**

Notes: Yields are measured in bushels per acre (bu/ac). Obs. denotes total county-year observations in each period (135 counties × years). IQR is the interquartile range ($Q_3 - Q_1$). CV is the coefficient of variation ($100 \times \sigma/\mu$). *Source:* Author's calculation on USDA NASS Quick Stats data (1951–2025).

Table 3.3. Statistical and diagnostic comparison of candidate detrending models across the 75-year panel (1951–2025).

Specification	Params (k)	R^2	Adj. R^2	RSE (bu/ac)	AIC	BIC	ADF p	OOS RMSE (bu/ac)
Panel A: Regional Aggregate Mean Yield Trajectory ($N = 75$)								
Linear OLS	2	0.9323	0.9313	2.87	372.7	377.4	< 0.001	**3.277**
Quadratic Polynomial	3	0.9398	0.9382	2.72	365.8	372.8	< 0.001	3.506
Cubic Polynomial	4	0.9428	0.9404	2.67	364.1	373.4	< 0.001	3.839
Panel B: County-Level Generalization (135 ***Counties*** × 30 ***Pre-Test Years*** = 4,050 ***Forecasts***)								
Linear OLS (County-Specific)	2	0.8169	0.8096	5.12	—	—	< 0.001	**5.767**
Quadratic (County-Specific)	3	0.8274	0.8196	4.98	—	—	< 0.001	6.145
Cubic (County-Specific)	4	0.8335	0.8239	4.92	—	—	< 0.001	6.618

Notes: k denotes parameter count. In-sample fit metrics (R^2, Adj. R^2, RSE, AIC, BIC, ADF p) are estimated across the complete 75-year historical panel (1951–2025, 10,125 county-year observations). OOS RMSE evaluates 1-step-ahead out-of-sample forecast accuracy across 30 expanding windows restricted to the pre-test period (1981–2010, 4,050 county-year forecasts, base training initiated on 1951–1980), strictly preventing data snooping into the primary machine-learning test set (2011–2025). In Panel B, BIC strictly favors Linear OLS in 62.2% of counties (84/135). Linear OLS minimizes out-of-sample forecast error, eliminating the endpoint instability and overfitting inherent in higher-order polynomial specifications. *Source:* Author's calculation on USDA NASS yield panel (1951–2025).

Table 3.3 evaluates alternative county-applicable detrending specifications. In Panel A (Regional Aggregate Mean, $N = 75$), Linear OLS accounts for over 93% of annual variation with two parameters ($k = 2, R^2 = 0.9323$). While quadratic and cubic polynomials marginally reduce in-sample residual standard error (reaching $R^2 = 0.9428$ for the cubic specification), they increase parameter complexity.

Panel B evaluates out-of-sample forecast accuracy across 4,050 expanding-window county-year evaluations restricted to the pre-test period (1981–2010), strictly preventing data snooping into the subsequent machine-learning test set (2011–2025). County-specific linear OLS achieves the lowest out-of-sample RMSE, at 5.767 bu/ac (compared to

6.145 bu/ac for quadratic and 6.618 bu/ac for cubic), and is preferred by BIC in 89/135 counties over quadratic (65.9%) and in 84/135 counties overall (62.2%). Higher-order polynomials increase forecast error outside the sample despite their closer in-sample fit. To respect temporal causality and prevent look-ahead bias, model selection must be anchored to this pre-test era, during which structural break tests show no parameter instability (Chow test $p = 0.093$). Full-sample properties across 1951–2025 (where a post-2014 technological shift emerges) and alternative flexible trends are examined in Appendix A as robustness stress tests. The Augmented Dickey-Fuller test firmly rejects the null hypothesis of a unit root in the detrended series ($p < 0.001$), confirming that the yield anomalies are stationary ($I(0)$).

3.2.2 Spatial Heterogeneity and Regional Productivity Divergence

While secular growth is positive throughout the Midwest, historical productivity distributions exhibit marked spatial heterogeneity across states (Figure 3.3). Baselines vary substantially in cross-sectional intercepts α_c: Iowa (historical mean 39.73 bu/ac) and Illinois (38.35 bu/ac) define the highest-yielding core, whereas Missouri records the lowest historical baseline (29.04 bu/ac). This pronounced cross-sectional spread formally justifies estimating individual fixed effects and county-specific detrending parameters, precluding a uniform pooled intercept.

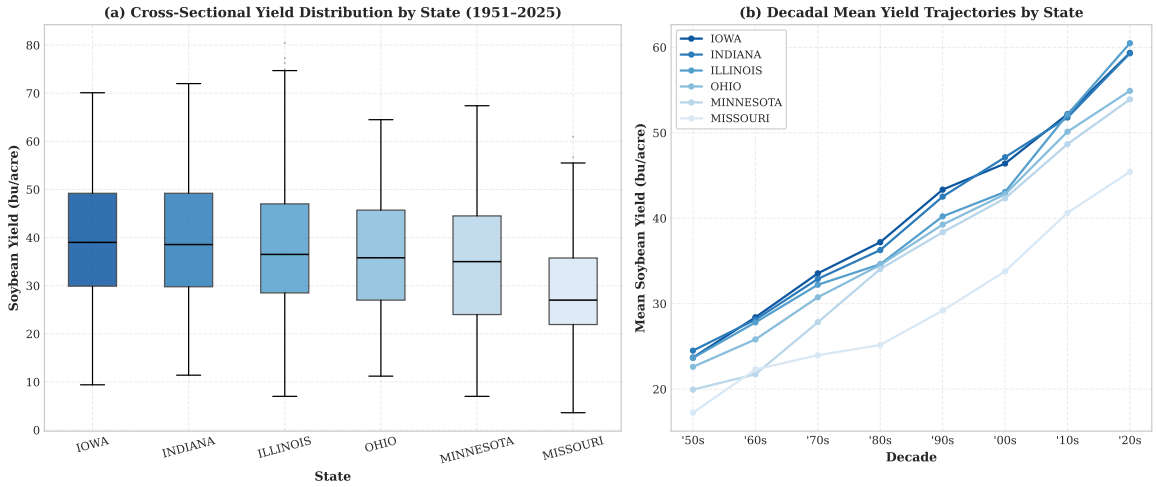


Figure 3.3. Spatial heterogeneity of county soybean yields: (a) cross-sectional yield distributions by state ordered by median yield; (b) decadal mean yield trajectories across the six Midwestern states from the 1950s to the 2020s.

Source: Author's calculation on USDA NASS data.

Furthermore, technological growth trajectories vary across regions: linear trend slopes β_c range from +0.509 bu/ac/year in Minnesota to +0.386 bu/ac/year in Missouri, confirming the statistical requirement to estimate heterogeneous slopes at county level. Finally, cross-county dispersion displays distinct regional patterns: Indiana and Ohio show narrow within-state variance around their historical means (39.52 and 36.88 bu/ac), whereas

Illinois and Minnesota exhibit wide internal variance, highlighting the need to evaluate forecasting accuracy across disaggregated local scales.

3.2.3 Macro-Climatic Historical Yield Shocks and Biological Asymmetry

The 75-year panel contains severe macro-climatic shock campaigns that serve as out-of-sample benchmark cohorts. Using an anomaly threshold of $|\epsilon_t/\hat{y}_t| \geq 10\%$, Table 3.4 classifies historical extreme campaigns into shortfall and bumper regimes.

The empirical distribution of yield anomalies exhibits pronounced **biological asymmetry** (Table 3.4). Favorable harvest campaigns cluster tightly near +10%, constrained by the biophysical ceiling of radiation use efficiency. In contrast, severe abiotic stressors possess no physiological floor short of complete failure, generating an elongated negative tail exceeding −25% with notable negative skewness (−1.12, Jarque-Bera $p < 0.001$).

3.3 Surface Weather Data: ECMWF ERA5-Land

3.3.1 Dataset Characteristics and Meteorological Ingestion

We derive meteorological inputs from ECMWF ERA5-Land, which provides continuous hourly surface weather fields at a $0.1° \times 0.1°$ resolution (≈ 9 km) from 1950 to 2025.

The hourly ECMWF ERA5-Land reanalysis provides a comprehensive suite of atmospheric and land-surface fields. From this broader catalogue, raw variables such as surface pressure (SP) and 10-meter horizontal wind vectors (u_{10}, v_{10}) serve strictly as intermediate thermodynamic inputs to compute standardized FAO-56 Penman-Monteith reference evapotranspiration (ET_0, Eq. (3.5)) and vapor pressure deficit (VPD, Eq. (3.4)). Deep soil horizons beyond mature soybean taproot depths ($SM_4 > 1$ m), soil temperatures, and snowpack dynamics are excluded a priori to prevent dimensionality inflation and avoid introducing uninformative predictors during the summer vegetative cycle.

Model inputs contain **17 daily agrometeorological indicators** (Table 3.5): nine primary surface fields and eight derived indicators. The derived set includes two binary heat-day indicators and the separate excess-temperature measure EDD_{30}.

3.3.2 Derived Agrometeorological Variables and Bioclimatic Indicators

Crop yields respond non-linearly to compound atmospheric-hydrological stress (Schlenker and Roberts, 2009). We therefore transform daily meteorological fields into targeted bioclimatic indicators (Table 3.5, Panel B).

Vapor Pressure Deficit (VPD). Atmospheric evaporative demand is parameterized via Vapor Pressure Deficit, which quantifies the drying power of the air. Saturation vapor pressure $e_s(T_{\text{mean}})$ and actual vapor pressure $e_a(T_{\text{dew}})$ (in kPa) are computed via Tetens

Table 3.4. Historical macro-climatic shock years, commodity price responses, and associated drivers (1951–2025).

Year	Yield Anom. (bu/ac)	Rel. Anom. (%)	Max Price Swing (%)	Primary Meteorological / Ecological Driver
Panel A: Major Adverse Macro-Climatic Yield Shocks (Negative Anomalies)				
1988	-9.64	-25.6%	+33.0%	Historic severe summer drought and persistent heatwave across the Corn Belt.
2003	-9.72	-21.6%	+59.5%	Late-summer heat and moisture stress compounded by widespread soybean aphid outbreak.
1974	-6.09	-19.7%	+58.6%	Excessively wet spring delaying planting, followed by midsummer drought and early frost.
2012	-5.80	-11.7%	+17.4%	Extensive summer flash drought and record July heat across the central United States.
1983	-4.02	-11.4%	+36.0%	Prolonged July-August heatwave and severe drought across the Corn and Soybean Belt.
1993	-4.22	-10.5%	+17.5%	Great Upper Mississippi River flood; prolonged waterlogging, root asphyxia, and disease.
Panel B: Prominent Favorable Bumper Harvests (Positive Anomalies)				
1952	+2.41	+11.9%	—	Exceptionally favorable vegetative and pod-filling moisture across the central belt.
2021	+5.79	+10.8%	-14.4%	Optimal reproductive rainfall and moderate temperatures throughout August.
2016	+5.53	+10.8%	+13.2%	Broadly distributed summer rains and minimal extreme heat stress across the Midwest.
1994	+4.33	+10.7%	-19.3%	Record-setting benign growing conditions with timely August rains and cool nights.

Notes: Yield Anom. $= y_t - \hat{y}_t$; Rel. Anom. $= (y_t - \hat{y}_t)/\hat{y}_t \times 100$, where $\hat{y}_t$ is the panel linear trend.

Max Price Swing: peak percentage departure in national commodity prices over the 12-month campaign window. *Source:* Author's calculation on USDA NASS and farmdoc (University of Illinois) data.

Table 3.5. ERA5-Land surface meteorological variables and derived daily agrometeorological indicators.

Category	Variable and Unit	Physical Description and Agronomic Rationale
Panel A: Primary ERA5-Land surface fields		
Thermal Dynamics	$T_{\text{mean}}, T_{\max}, T_{\min}$ (°C)	Daily mean, maximum, and minimum 2 m air temperature; thermal accumulation and heat stress.
	T_{dew} (°C)	Daily mean 2 m dewpoint temperature; near-surface moisture baseline.
Hydrological Fluxes	P (mm/d)	Total daily precipitation; surface water supply, recharge, and waterlogging.
Radiative Forcing	$SSRD$ (MJ/m^2/d)	Surface downward solar radiation; photosynthetically active radiation driving biomass assimilation.
Pedological Moisture	SM_1, SM_2, SM_3 (m^3/m^3)	Volumetric soil water across depths: 0–7 cm (seedbed), 7–28 cm (upper root zone), and 28–100 cm (deep root reservoir).
Panel B: Derived Agrometeorological and Bioclimatic Indicators		
Evaporative Demand	VPD (kPa)	Vapor pressure deficit $e_s(T_{\text{mean}}) - e_a(T_{\text{dew}})$; atmospheric drying power driving stomatal closure.
Crop Water Demand	ET_0 (mm/d)	FAO-56 Penman-Monteith reference evapotranspiration (Allen et al., 1998).
Climatic Balance	$P - ET_0$ (mm/d)	Climatic water balance; net daily atmospheric moisture deficit or surplus.
Thermal Pacing	GDD (°C · d)	Growing Degree Days (10–30°C); vegetative and reproductive phenological progression pacing.
Extreme Heat Stress	HD_{30}, HD_{35} (binary)	Daily heat day exceedances $\mathbb{I}(T_{\max} \geq 30°\text{C})$ and $\geq 35°$C (Schlenker and Roberts, 2009).
Heat Intensity	EDD_{30} (°C · d)	Daily excess temperature $\max(0, T_{\max} - 30)$; thermal damage accumulation.
Root-Zone Reservoir	SM_{root} (m^3/m^3)	Depth-weighted soil water column (0–100 cm, Eq. (3.10)); multi-month hydrological buffer.

Source: Author's elaboration on ECMWF ERA5-Land reanalysis specifications and agronomic literature (Allen et al., 1998; Schlenker and Roberts, 2009).

formulations:

$$e_s(T_{\text{mean}}) = 0.61078 \exp\left(\frac{17.27\,T_{\text{mean}}}{T_{\text{mean}} + 237.3}\right), \quad e_a(T_{\text{dew}}) = 0.61078 \exp\left(\frac{17.27\,T_{\text{dew}}}{T_{\text{dew}} + 237.3}\right) \tag{3.3}$$

$$VPD = \max\left(0,\, e_s(T_{\text{mean}}) - e_a(T_{\text{dew}})\right) \tag{3.4}$$

Elevated VPD (> 1.5 kPa) triggers stomatal closure to prevent dehydration, suppressing transpirational cooling and carbon assimilation during reproductive stages.

Reference Evapotranspiration (ET_0) and Climatic Water Balance ($P - ET_0$). We model potential atmospheric water demand via the standardized FAO-56 Penman-Monteith equation for a short grass reference canopy:

$$ET_0 = \frac{0.408\,\Delta\,(R_n - G) + \gamma\,\frac{900}{T_{\text{mean}}+273}\,u_2\,(e_s - e_a)}{\Delta + \gamma\,(1 + 0.34\,u_2)} \tag{3.5}$$

where R_n is net surface radiation (MJ/m^2/day), $G \approx 0$ is daily soil heat flux density, u_2 is 2m wind speed (m/s), Δ is the saturation vapor pressure slope, and γ is the psychrometric constant. Daily climatic water balance tracks atmospheric moisture deficits or recharge surpluses:

$$P_{\text{net}} = P - ET_0 \tag{3.6}$$

Thermal Pacing and Extreme Heat Exposure ($GDD, HD_{30}, HD_{35}, EDD_{30}$). We track phenological progression through Growing Degree Days (GDD, base 10°C, cap 30°C):

$$GDD = \max\left(0,\, \frac{\min(T_{\max}, 30) + \max(T_{\min}, 10)}{2} - 10\right) \tag{3.7}$$

To capture non-linear thermal damage thresholds (Schlenker and Roberts, 2009), we quantify extreme heat exposure via daily binary exceedance counts:

$$HD_{30} = \mathbb{I}(T_{\max} \geq 30°\text{C}), \quad HD_{35} = \mathbb{I}(T_{\max} \geq 35°\text{C}) \tag{3.8}$$

In addition to frequency counts, thermal damage intensity is captured by Extreme Degree Days accumulating degrees above the 30°C threshold:

$$EDD_{30} = \max\left(0,\, T_{\max} - 30\right) \tag{3.9}$$

Integrated Root-Zone Soil Moisture (SM_{root}). We aggregate volumetric soil moisture across discrete depth horizons into an integrated column accessible to mature soybean

taproots (0–100 cm):

$$SM_{\text{root}} = 0.07\,SM_1 + 0.21\,SM_2 + 0.72\,SM_3 \tag{3.10}$$

weighting by layer thickness (7 cm, 21 cm, 72 cm). This metric buffers transient surface dry spells and tracks subsoil moisture memory sustaining transpirational uptake during summer rainfall deficits (Vijverberg et al., 2023).

3.3.3 Spatial Aggregation: Exact Planar Polygon Intersection

To map gridded weather fields into administrative county polygons, we compute exact planar polygon intersections (`spatial_weights.csv`). Across the 135 study counties and the ERA5-Land 0.1° grid mesh, we identify 3,344 unique intersection polygons with normalized area weights:

$$w_{c,g} = \frac{\text{Area}(c \cap g)}{\text{Area}(c)}, \quad \sum_{g \in \mathcal{G}_c} w_{c,g} = 1 \tag{3.11}$$

where $\text{Area}(c \cap g)$ is the planar intersection area between county polygon c and grid cell g, and $\mathcal{G}_c$ denotes the set of intersecting grid cells.

As illustrated in Figure 3.4, area-weighted polygon integration preserves the continuous precipitation volume, radiative flux, and atmospheric mass within each county polygon without geometric distortion.

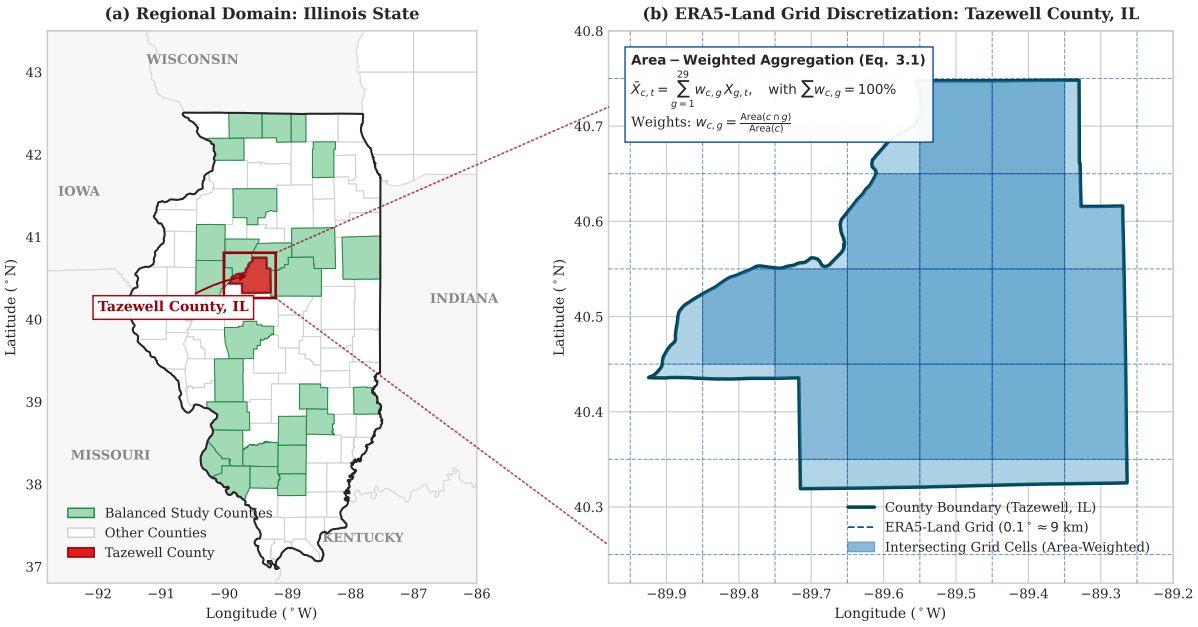


Figure 3.4. Spatial grid architecture and area-weighted discretization of ERA5-Land weather data: (a) regional study domain across Illinois ($N = 28$ counties); (b) overlay of regular $0.1° \times 0.1°$ (≈ 9 km) grid cells and area-weighted planar intersection for Tazewell County.

Source: Author's elaboration on ECMWF ERA5-Land grid and US Census TIGER/Line county boundaries.

3.3.4 Temporal Aggregation and Crop Campaign Structure

We spatially integrate hourly ERA5-Land fields across county polygons via Equation (3.11) and aggregate them into daily meteorological series. For crop year Y, model inputs cover November 1 of $Y-1$ through October 31 of Y, comprising 365 or 366 days. Each fitted horizon uses only the dates observed through its month-end cutoff. Calendar-month tabular aggregation and continuous recurrent windows both retain all available days.

3.4 Exploratory Agro-Climatic Analysis and Yield Sensitivity

We examine seasonal weather trajectories, spatial contrasts, and monthly associations with detrended yield anomalies across the 75-year panel.

3.4.1 Intra-Seasonal Agro-Climatic Trajectories: Cumulative and Dynamic Shock Archetypes

Cumulative exposures and daily environmental states describe complementary dimensions of crop stress. Figure 3.5 traces seasonal thermal and hydrological accumulation from April through October. Figure 3.6 shows 15-day centred rolling profiles of VPD, root-zone soil moisture, and maximum temperature. These descriptive profiles contrast the 2012 drought and 1993 flood with the 2016 and 1994 bumper seasons against the 75-year climatological envelope.

The empirical trajectories in Figures 3.5–3.7 establish three statistical properties of agro-climatic shocks:

1. **Non-Linearity and Concavity of Hydrological Response:** The 2012 drought (-11.7%) and the 1993 flood (-10.5%) produced nearly identical yield collapses despite opposite hydrological extremes. In 2012, cumulative water balance $\sum(P - ET_0)$ plunged to -300 mm alongside an acute spike in heat days, whereas in 1993, excessive rainfall generated a surplus of $+400$ mm with zero extreme heat. This statistical symmetry confirms that crop response to water availability is non-linear and concave: a standard monotonic linear model cannot penalize both severe deficits and extreme waterlogging, motivating non-parametric tree ensembles and threshold features.

2. **Compound Shock Dynamics and Multicollinearity:** In 2012, atmospheric evaporative demand and thermal stress surged simultaneously: rolling VPD exceeded 1.8 kPa, $T_{\max}$ exceeded 33°C, and root-zone soil moisture dropped below 0.18 m^3/m^3 for four consecutive weeks across central Iowa and Illinois (Figure 3.6). This persistent co-occurrence illustrates high multivariate collinearity among climate predictors, requiring regularized shrinkage (ElasticNet) and sequential tracking (LSTM).

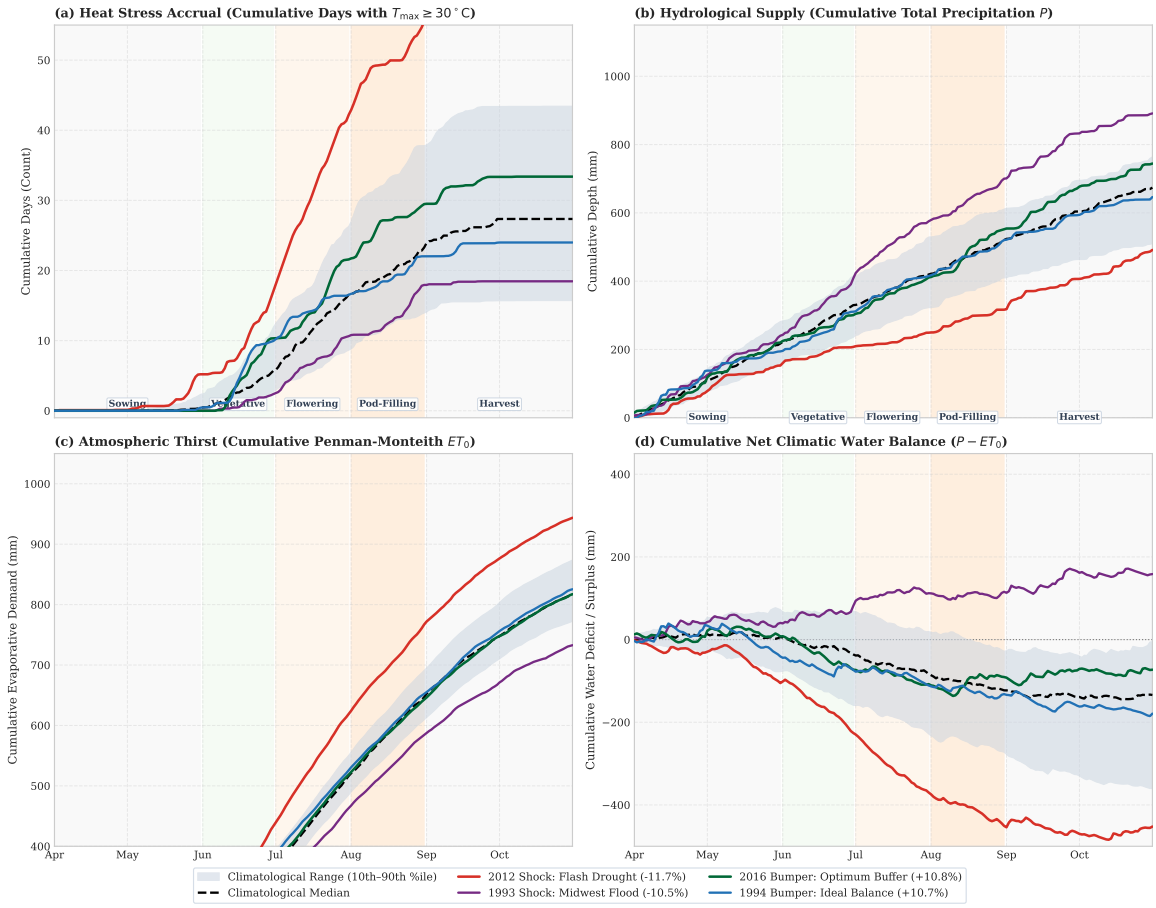


Figure 3.5. Cumulative intra-seasonal trajectories of heat stress ($\sum HD_{30}$), precipitation ($\sum P$), evapotranspiration ($\sum ET_0$), and climatic water balance ($\sum(P - ET_0)$) across developmental stages (April–October) for benchmark shortfall (1993, 2012) and bumper (1994, 2016) campaigns relative to the climatological envelope.

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data.

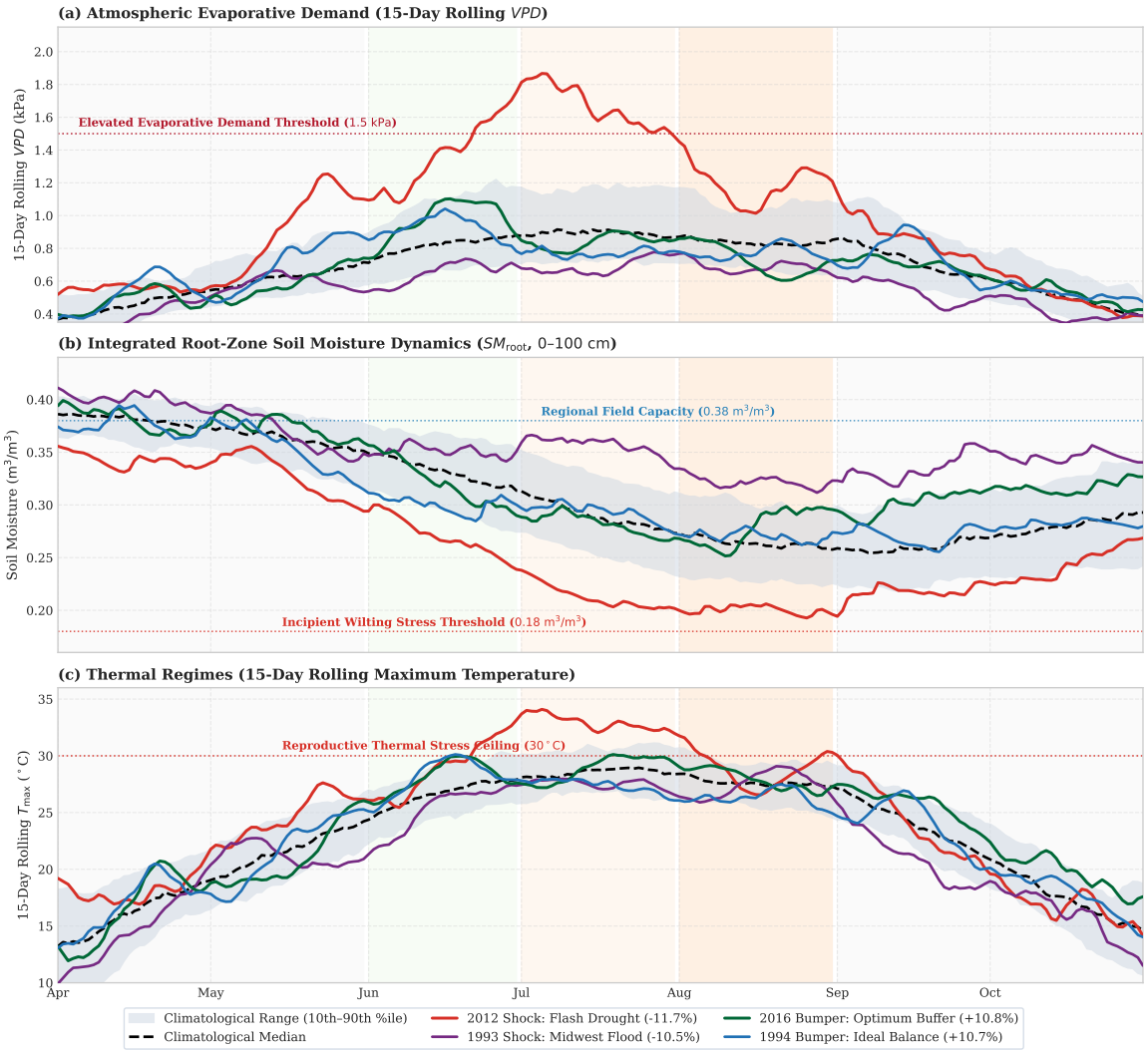


Figure 3.6. Daily 15-day rolling agro-climatic state profiles (VPD, root-zone soil moisture SM_{root}, and $T_{\max}$) across the growing cycle (April–October) for benchmark shortfall and bumper campaigns against the 75-year climatological envelope, highlighting reproductive stress thresholds (July–August).

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data.

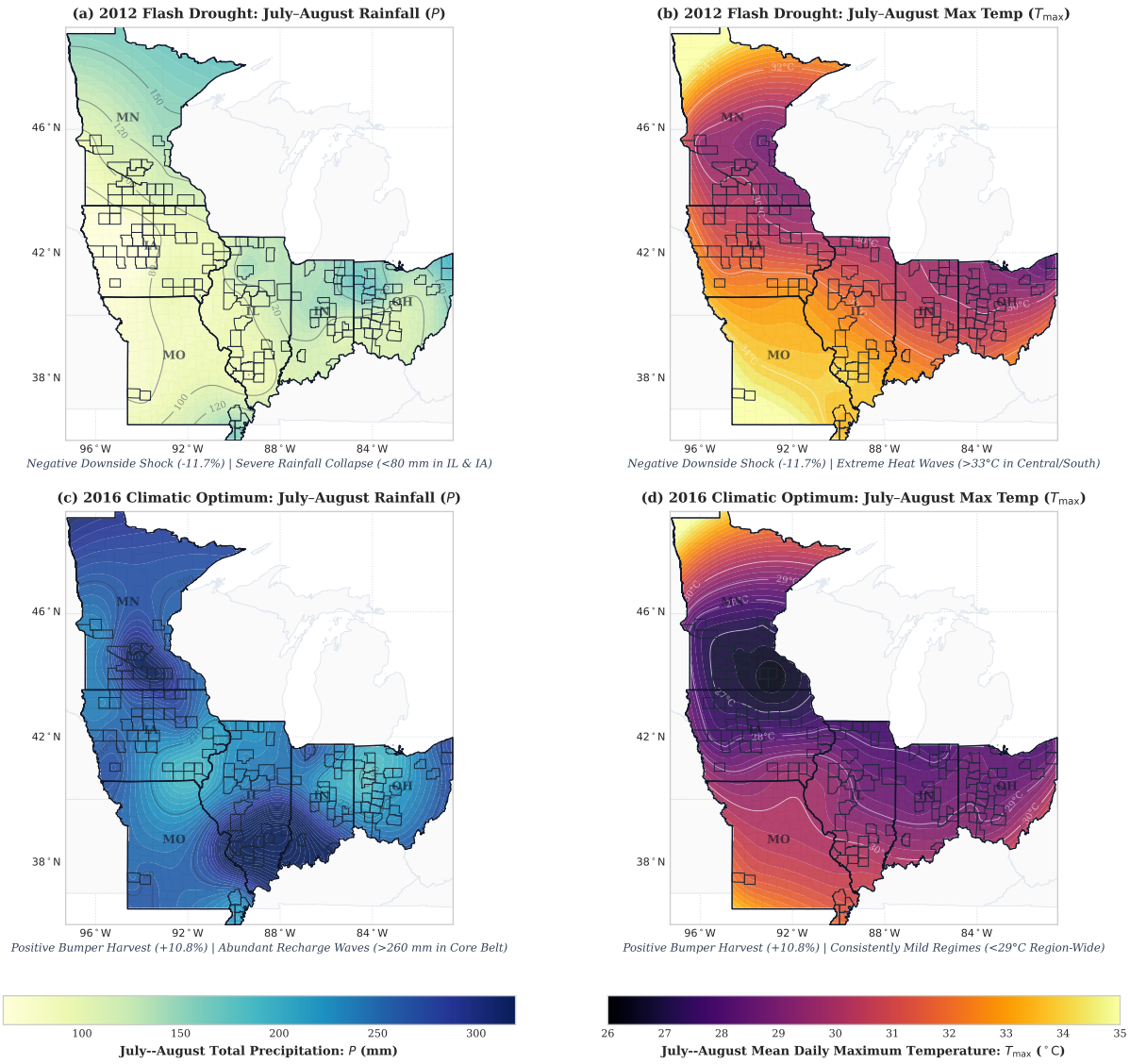


Figure 3.7. Spatial distribution of July–August cumulative precipitation (P) and mean maximum temperature ($T_{\max}$) across the 135 study counties, contrasting the 2012 midsummer flash drought against the 2016 bumper harvest.

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data.

3. **Synoptic Regional Scale and Spatial Autocorrelation:** Figure 3.7 shows that the 2012 flash drought affected a contiguous multi-state corridor (Illinois, Indiana, Missouri, and southern Iowa), whereas Minnesota was largely spared. The broad spatial footprint of atmospheric anomalies induces strong cross-county correlation, requiring expanding-window cross-validation over complete calendar years to prevent spatial information leakage.

3.4.2 Phenological Climate-Yield Sensitivity and Lead-Time Dynamics

To quantify how the statistical dependence between atmospheric forcing and final yield evolves across the crop lifecycle, Figure 3.8 presents bivariate Pearson correlation coefficients (r) between detrended county yield anomalies ($\epsilon_{c,t}$) and monthly atmospheric anomalies across all 135 study counties over 75 years.

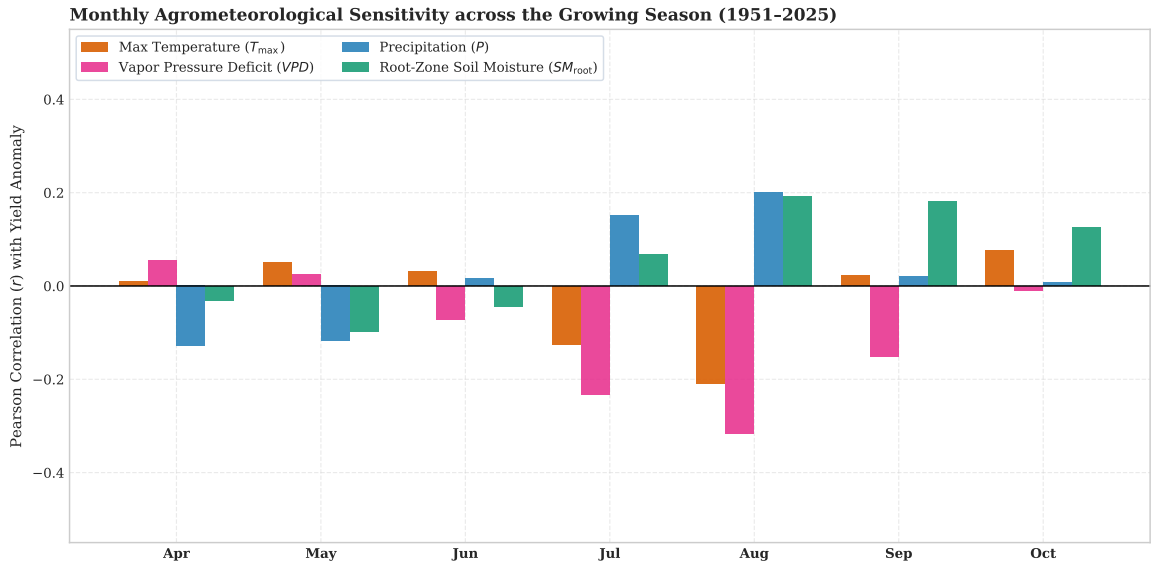


Figure 3.8. Cross-sectional Pearson correlation coefficients (r) between detrended county soybean yield anomalies ($\epsilon_{c,t}$) and monthly agro-climatic anomalies ($T_{\max}$, VPD, P, SM_{root}) across the crop cycle (April–October, 1951–2025).

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data.

The correlation structure exhibits marked seasonal heteroskedasticity. During preplanting and emergence (April–May), correlation coefficients remain weak and close to zero ($|r| < 0.10$). Predictive signal concentrates heavily during midsummer reproductive growth (July–August), where correlations reach $r \approx -0.45$ for VPD and $T_{\max}$, and $r \approx +0.40$ for precipitation and root-zone soil moisture. As crops reach physiological maturity in September, correlations weaken sharply.

While these bivariate profiles demonstrate the concentration of climate sensitivity during reproductive growth, pairwise correlations capture only marginal associations and ignore compound multivariate interactions. The convergence of hydrological concavity (Figure 3.5), persistent stress duration (Figure 3.6), synoptic spatial coherence (Figure 3.7),

and phenological concentration (Figure 3.8) provides the empirical foundation for the multi-horizon predictive framework formalized in Chapter 4.

4. Methodology

Building on the 75-year panel characterized in Chapter 3, this chapter formalizes the statistical learning framework for multi-horizon soybean yield anomaly forecasting across the Midwestern Corn Belt. The predictive architecture pursues two objectives: determining the operational lead time where meteorological predictors yield significant skill over the naive trend benchmark, and evaluating whether non-linear algorithmic complexity improves performance under strict out-of-sample expanding-window validation.

4.1 Problem Formulation and Multi-Horizon Forecast Setup

4.1.1 Continuous Target Formulation and Train-Only Detrending

Realized county soybean yields decompose additively into a secular technological trend and an annual climatic anomaly (Section 3.2.1):

$$y_{c,Y} = \tau_{c,Y} + \epsilon_{c,Y} \tag{4.1}$$

where $c \in \{1, \ldots, C\}$ is the county index ($C = 135$), $Y \in \{1951, \ldots, 2025\}$ denotes harvest crop year, $\tau_{c,Y}$ designates the secular technological trend, and $\epsilon_{c,Y}$ isolates the weather-induced anomaly. Predictive models target the continuous anomaly $\epsilon_{c,Y} \in \mathbb{R}$ directly.

Detrending parameters are estimated on training observations to prevent look-ahead leakage. For harvest year Y, county-specific linear trend parameters $(\hat{\alpha}_c^{(Y)}, \hat{\beta}_c^{(Y)})$ are fitted over historical campaigns $\mathcal{T}_{\text{train}}(Y) = \{1951, \ldots, Y-1\}$ by minimizing the mean squared error:

$$\hat{\tau}_{c,Y} = \hat{\alpha}_c^{(Y)} + \hat{\beta}_c^{(Y)} Y, \quad (\hat{\alpha}_c^{(Y)}, \hat{\beta}_c^{(Y)}) = \arg\min_{\alpha,\beta} \sum_{t \in \mathcal{T}_{\text{train}}(Y)} \left(y_{c,t} - (\alpha + \beta t)\right)^2 \tag{4.2}$$

Training targets are $\epsilon_{c,t} = y_{c,t} - \hat{\tau}_{c,t}$. On held-out evaluation year Y, models predict continuous anomaly $\hat{\epsilon}_{c,Y}$, and total yield levels are reconstructed post-hoc:

$$\hat{y}_{c,Y} = \hat{\tau}_{c,Y} + \hat{\epsilon}_{c,Y} \tag{4.3}$$

4.1.2 Daily Weather Aggregation and Feature Dimensionality

The prediction target is a single annual scalar per county ($\epsilon_{c,Y} \in \mathbb{R}$), while meteorological forcing operates daily across $K = 17$ indicators. Ingesting unconstrained daily weather across 365 days generates approximately 6,205 features per county-year, inducing severe multicollinearity and variance inflation.

To preserve phenological timing while restricting dimensionality, daily weather fields are aggregated into structured monthly blocks:

$$w_{k,j} = \begin{cases} \sum_{d \in \mathcal{D}_j} v_{k,d}, & \text{for cumulative flux indicators } k \in \mathcal{K}_{\text{flux}} \\ \frac{1}{|\mathcal{D}_j|} \sum_{d \in \mathcal{D}_j} v_{k,d}, & \text{for environmental state indicators } k \in \mathcal{K}_{\text{state}} \end{cases} \tag{4.4}$$

where $\mathcal{K}_{\text{flux}} = \{P, SSRD, ET_0, P - ET_0, GDD, HD_{30}, HD_{35}, EDD_{30}\}$ denotes cumulative exposures, $\mathcal{K}_{\text{state}} = \{T_{\text{mean}}, T_{\max}, T_{\min}, T_{\text{dew}}, SM_1, SM_2, SM_3, VPD, SM_{\text{root}}\}$ designates environmental states, and $\mathcal{D}_j$ denotes calendar days in campaign month $j \in \{1, \ldots, 12\}$. For each elapsed month j, this defines a 17-dimensional meteorological summary vector $\mathbf{w}_{c,Y,j} \in \mathbb{R}^{17}$. Monthly aggregation preserves dimensional stability ($P_H \leq 206$) while aligning features with broader crop developmental phases (Hoffman et al., 2020). Sub-monthly tabular concatenation would induce severe multicollinearity on historical samples (Sharma et al., 2025); high-frequency decad dynamics are evaluated separately through sequential deep architectures (Section 4.3.4).

4.1.3 Crop Campaign Timeline and Multi-Horizon Feature Expansion

The crop campaign spans November 1 of year $Y-1$ to October 31 of harvest year Y. The pre-season origin at October 31 of $Y-1$ ($H = 12$) ingests zero weather predictors ($m = 0, P_{\text{met}} = 0$), serving as an uninformed benchmark that outputs a zero anomaly ($\hat{\epsilon}_{c,Y}^{(12)} = 0$) and relies entirely on the trend projection ($\hat{y}_{c,Y}^{(12)} = \hat{\tau}_{c,Y}$). Operational horizons $H \in \{11, \ldots, 1\}$ correspond to monthly cutoffs from December 31 of $Y-1$ through October 31 of Y, ingesting $m = 13 - H \in \{2, \ldots, 12\}$ elapsed campaign months.

At forecast horizon H, the pure meteorological feature vector $\mathbf{x}_{c,Y}^{(H)}$ concatenates normalized spatial coordinates ($\tilde{\text{Lat}}_c, \tilde{\text{Lon}}_c$) (Section 4.2) with elapsed monthly weather summaries:

$$\mathbf{x}_{c,Y}^{(H)} = \left[\tilde{\text{Lat}}_c,\ \tilde{\text{Lon}}_c,\ \mathbf{w}_{c,Y,1}^T,\ \ldots,\ \mathbf{w}_{c,Y,m}^T \right]^T \in \mathbb{R}^{P_H} \tag{4.5}$$

To evaluate whether past yield persistence provides incremental skill over current weather conditions, we specify an integrated feature vector that appends the prior-year detrended yield anomaly ($\epsilon_{c,Y-1}$):

$$\mathbf{x}_{c,Y}^{(H,\text{int})} = \left[\mathbf{x}_{c,Y}^{(H)T},\ \epsilon_{c,Y-1} \right]^T \in \mathbb{R}^{P_H+1} \tag{4.6}$$

where $\epsilon_{c,Y-1} = y_{c,Y-1} - \hat{\tau}_{c,Y-1}$ designates the prior-year detrended yield anomaly estimated strictly within training partition $\mathcal{T}_{\text{train}}(Y)$. Training-only detrending removes the estimated secular component from this predictor. The lag ablation measures its incremental predictive contribution; it does not identify a biophysical carryover mechanism.

Feature dimensionality expands as meteorological records accumulate across the grow-

ing season:

$$P_H = \begin{cases} P_{\text{geo}} = 2, & \text{for } H = 12 \quad (m = 0, \text{ zero weather}) \\ P_{\text{geo}} + K \cdot m = 2 + 17 \cdot m, & \text{for } H \in \{11, \ldots, 1\} \quad (m = 2, \ldots, 12) \end{cases} \tag{4.7}$$

This schedule structures information accumulation from $P_{11} = 36$ predictors at December 31 through $P_1 = 206$ in October (or $P_1 = 207$ with prior-year carryover), synchronized to the state-specific harvest closure horizon $H^*(s)$:

$$H^*(s) = \begin{cases} 2 \quad (\text{September 30}), & \text{for Minnesota} \\ 1 \quad (\text{October 31}), & \text{for IA, IL, IN, MO, OH} \end{cases} \tag{4.8}$$

Across all preliminary horizons $H \in \{12, \ldots, 2\}$, meteorological features are aligned to identical calendar month-ends across all six states (October 31 of $Y-1$ through September 30 of harvest year Y). The state-specific closure rule $H^*(s)$ applies exclusively at the final pre-harvest horizon: at October 31 ($H = 1$), forecasts for Minnesota are frozen at their September 30 state ($H = 2$) to eliminate post-harvest weather noise from already harvested fields, while models for central and southern states ingest October weather.

4.1.4 Direct Horizon-Specific Modeling Paradigm

We implement direct multi-horizon forecasting. For each lead time $H \in \{12, \ldots, 1\}$, an independent model $\mathcal{M}_H$ is trained on horizon-specific feature vector $\mathbf{x}_{c,Y}^{(H)}$:

$$\hat{\epsilon}_{c,Y}^{(H)} = \mathcal{M}_H \left(\mathbf{x}_{c,Y}^{(H)}; \boldsymbol{\Theta}_H \right) \tag{4.9}$$

where $\boldsymbol{\Theta}_H$ denotes the parameter vector. Direct forecasting prevents compounding error propagation from iterated rollouts. It allows model weights to reconfigure across developmental stages, prioritizing early soil moisture at long lead times and reproductive heat stress during late-summer origins.

4.1.5 Pooled Panel Estimation and County-Level Evaluation

Models are estimated within a pooled cross-sectional panel framework across all $C = 135$ counties simultaneously. Separate county-by-county estimation with 60 to 74 historical observations causes parameter collapse ($P_H > T$) against feature vectors with up to 206 variables. Pooling expands sample size to $N_{\text{train}} = 135 \times |\mathcal{T}_{\text{train}}(Y)|$ observations. The initial training partition (1951–2010) provides 8,100 county-years, expanding to 9,990 observations at the final 2025 fold.

While estimation parameters are pooled, predictions remain county-specific: for each county c and year Y, the model generates an individual anomaly forecast $\hat{\epsilon}_{c,Y}^{(H)}$ by evaluating localized weather series and spatial coordinates $(\tilde{\text{Lat}}_c, \tilde{\text{Lon}}_c)$. County yield forecasts

are reconstructed using county-specific trends $\hat{\tau}_{c,Y}$. Empirical skill is evaluated both across the pooled panel and via annual regional averages (Section 4.5).

4.2 Spatial Encoding: Continuous Centroid Coordinates

Spatial location is encoded exclusively through normalized continuous centroid coordinates:

$$\mathbf{s}_c = (\mathrm{Lat}_c, \mathrm{Lon}_c) \in \mathbb{R}^2, \quad P_{\mathrm{geo}} = 2 \tag{4.10}$$

Coordinates are scaled to $[0, 1]$ across the geographic domain of the 135 study counties:

$$\tilde{\mathrm{Lat}}_c = \frac{\mathrm{Lat}_c - \min_i \mathrm{Lat}_i}{\max_i \mathrm{Lat}_i - \min_i \mathrm{Lat}_i}, \quad \tilde{\mathrm{Lon}}_c = \frac{\mathrm{Lon}_c - \min_i \mathrm{Lon}_i}{\max_i \mathrm{Lon}_i - \min_i \mathrm{Lon}_i} \tag{4.11}$$

where $\min_i$ and $\max_i$ define the spatial boundaries of the panel domain. This representation captures continuous geographic gradients without imposing artificial boundary discontinuities at state borders, enabling models to condition environmental response functions on spatial location.

4.3 Candidate Forecasting Models and Complexity Ladder

We compare a deterministic trend benchmark, regularized linear regression, tree ensembles, and recurrent networks to evaluate the contribution of weather inputs and model complexity.

4.3.1 Reference Baseline: Naive Deterministic Trend Benchmark

The deterministic trend benchmark predicts a zero anomaly:

$$\hat{\epsilon}_{c,Y} = 0 \iff \hat{y}_{c,Y} = \hat{\tau}_{c,Y} \tag{4.12}$$

This benchmark extrapolates the county trend fitted on training data. It defines $H = 12$ and the reference value $R^2_{\mathrm{OOS}} = 0$. Positive R^2_{OOS} values at subsequent horizons indicate that a weather-augmented model achieves a lower mean squared prediction error than this naive benchmark. Whether this error reduction is statistically distinguishable from zero is evaluated through Diebold–Mariano tests (Section 4.5.3).

4.3.2 Regularized Linear Model: ElasticNet

ElasticNet minimizes a penalized sum of squared residuals combining L_1 (Lasso) and L_2 (Ridge) penalties:

$$\min_{\beta,\beta_0} \frac{1}{2N} \sum_{i=1}^{N} \left(\epsilon_i - (\beta_0 + \mathbf{x}_i^T \beta)\right)^2 + \lambda \left(\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2\right) \tag{4.13}$$

where $\lambda > 0$ controls shrinkage intensity and $\alpha \in [0, 1]$ governs penalty mixing. In agricultural weather panels, summer atmospheric variables exhibit substantial mutual collinearity. Pure Lasso (L_1) penalization tends to select an arbitrary single predictor from a correlated cluster, discarding related environmental signals, while pure Ridge (L_2) shrinks coefficients smoothly without eliminating uninformative winter weather noise. Elastic-Net combines L_1 sparsity for automated feature selection with the L_2 grouping property, stably retaining clusters of correlated agrometeorological indicators (James et al., 2023).

4.3.3 Decision Tree Ensembles: Random Forest and XGBoost

We specify two complementary tree ensembles to evaluate non-linear partitioning:

1. **Random Forest (Bagging):** Averages B deep regression trees trained on bootstrap samples (Sharma et al., 2025):

$$\hat{\epsilon}_{c,Y} = \frac{1}{B} \sum_{b=1}^{B} T_b \left(\mathbf{x}_{c,Y}; \boldsymbol{\Theta}_b \right) \tag{4.14}$$

   At each candidate split, candidate partitioning minimizes node variance over a random feature subset $p_{\text{split}} \leq P_H$. Random feature subsampling decorrelates individual trees, reducing ensemble variance on high-dimensional weather inputs without requiring complex step-size tuning (James et al., 2023).

2. **XGBoost (Gradient Tree Boosting):** Fits an additive sequence of M shallow regression trees minimizing a regularized loss (Torsoni et al., 2023):

$$\mathcal{L}^{(m)} = \sum_{i=1}^{N} \left[g_i f_m(\mathbf{x}_i) + \frac{1}{2} h_i f_m(\mathbf{x}_i)^2 \right] + \gamma_{\text{split}} T_{\text{leaves}} + \frac{1}{2} \lambda_{\text{reg}} \sum_{j=1}^{T_{\text{leaves}}} w_j^2 + \alpha_{\text{reg}} \sum_{j=1}^{T_{\text{leaves}}} |w_j| \tag{4.15}$$

   where g_i and h_i are first and second order loss gradients, T_{leaves} is leaf count, and w_j are leaf weights. Leaf complexity is constrained by split threshold γ_{split}, L_2 regularization λ_{reg}, and L_1 penalty α_{reg} (Géron, 2025). Column subsampling randomly restricts predictors at each tree level to prevent dominance by individual heat-stress indicators (James et al., 2023). Sequential updates employ shrinkage rate $\eta \in (0, 1]$:

$$\hat{\epsilon}_i^{(m)} = \hat{\epsilon}_i^{(m-1)} + \eta f_m(\mathbf{x}_i) \tag{4.16}$$

   Sequential gradient boosting evaluates whether regularized additive trees can isolate non-linear compound heat-moisture stress thresholds that linear models smooth over.

4.3.4 Sequential Deep Learning: Long Short-Term Memory Network

While tabular algorithms require flattened two-dimensional design matrices $\mathbf{x}_{c,Y}^{(H)} \in \mathbb{R}^{P_H}$ where temporal blocks expand horizontally across columns, sequential deep architectures decouple input feature dimensionality from temporal sequence length (Khaki et al., 2020).

The Long Short-Term Memory (LSTM) network operates directly on a three-dimensional sequence tensor:

$$\mathbf{X}^{(H)} \in \mathbb{R}^{N_{\text{train}} \times T_H \times D_{\text{in}}} \tag{4.17}$$

Here, N_{train} represents sample size, T_H denotes sequence length across elapsed campaign steps at horizon H, and $D_{\text{in}} = 19$ represents input channel dimensionality. Following the spatial encoding evaluation in Section 4.2, D_{in} comprises the $K = 17$ agrometeorological indicators concatenated with the two normalized continuous spatial coordinates $(\tilde{\text{Lat}}_c, \tilde{\text{Lon}}_c)$ ingested at every sequence step. This enables recurrent memory cells to modulate hydrothermal responses conditionally on geographical latitude and longitude without requiring discrete county embeddings.

Meteorological series are partitioned into consecutive 10-day windows beginning on November 1 of year $Y - 1$. Windows can cross calendar-month boundaries. At each horizon, the final window ends at the forecast cutoff and retains any remaining days. The full campaign contains 37 steps: 36 complete windows and a final window of five or six days. Leap days are retained. Sums use the actual available days, and state variables are averaged over the same dates. Batch tensors use masked trailing padding when campaign lengths differ; padded positions are excluded from scaling and attention.

The 10-day decad frequency was selected on historical validation records (1985–1995) against alternative 5-day and 30-day aggregations, optimizing the trade-off between resolving intra-monthly heat-stress persistence and preserving gradient stability, and fixed prior to evaluating the out-of-sample test partition.

At each sequence step $t \in \{1, \ldots, T_H\}$, the recurrent cell ingests the input vector $\mathbf{z}_t \in \mathbb{R}^{19}$ and updates its internal state via gated recurrent transitions:

$$\mathbf{f}_t = \sigma\left(\mathbf{W}_f \mathbf{z}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f\right) \tag{4.18}$$

$$\mathbf{i}_t = \sigma\left(\mathbf{W}_i \mathbf{z}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i\right) \tag{4.19}$$

$$\tilde{\mathbf{c}}_t = \tanh\left(\mathbf{W}_c \mathbf{z}_t + \mathbf{U}_c \mathbf{h}_{t-1} + \mathbf{b}_c\right) \tag{4.20}$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \tag{4.21}$$

$$\mathbf{o}_t = \sigma\left(\mathbf{W}_o \mathbf{z}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{b}_o\right) \tag{4.22}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh\left(\mathbf{c}_t\right) \tag{4.23}$$

where $\mathbf{h}_t \in \mathbb{R}^{d_h}$ denotes the hidden state vector, $\mathbf{c}_t \in \mathbb{R}^{d_h}$ represents the internal cell state, and $\sigma(\cdot)$ is the sigmoid activation function. Forget gate $\mathbf{f}_t$ regulates historical information retention; input gate $\mathbf{i}_t$ modulates candidate cell updates $\tilde{\mathbf{c}}_t$; and output gate $\mathbf{o}_t$ filters the

emitted hidden representation.

To synthesize information across all elapsed sequence steps without compressing the entire growing season into the single final state $\mathbf{h}_{T_H}$, we implement a Bahdanau temporal attention layer:

$$e_t = \mathbf{v}_a^T \tanh\left(\mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a\right) \tag{4.24}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T_H} \exp(e_k)}, \quad \sum_{t=1}^{T_H} \alpha_t = 1 \tag{4.25}$$

$$\mathbf{c}_{\text{ctx}} = \sum_{t=1}^{T_H} \alpha_t \mathbf{h}_t \in \mathbb{R}^{d_h} \tag{4.26}$$

where $\mathbf{W}_a \in \mathbb{R}^{d_a \times d_h}$ and $\mathbf{v}_a \in \mathbb{R}^{d_a}$ parameterize the attention projection ($d_a = 32$). Attention weights α_t quantify the relative contribution of each 10-day window, providing physiological transparency by mapping predictive focus across vegetative and reproductive growth stages.

We evaluate two projection head architectures mapping context vector $\mathbf{c}_{\text{ctx}}$ to scalar anomaly prediction $\hat{\epsilon}_{c,Y}^{(H)}$:

1. **Non-Linear PReLU Head (Weather-Only Specification):**

$$\hat{\epsilon}_{c,Y}^{(H)} = \mathbf{W}_2 \,\text{PReLU}\left(\mathbf{W}_1 \mathbf{c}_{\text{ctx}} + \mathbf{b}_1\right) + b_2 \tag{4.27}$$

   where $\text{PReLU}(z) = \max(0, z) + a \min(0, z)$ learns its negative activation slope during backpropagation. The selected weather-only specification has two recurrent layers with 128 hidden units, dropout of 0.5, and batches of 25. Coordinates enter at each sequence step; they are not appended again to the context vector.

2. **Linear Projection Head (Parsimonious and Integrated Specification):**

$$\hat{\epsilon}_{c,Y}^{(H)} = \mathbf{W}_{\text{out}} \mathbf{c}_{\text{ctx}} + b_{\text{out}} \tag{4.28}$$

   The linear specification uses two recurrent layers with 128 hidden units, dropout of 0.4, and batches of 25. Its integrated version appends lagged yield to form $[\mathbf{c}_{\text{ctx}}^T, \epsilon_{c,Y-1}]^T$. The lag therefore enters the prediction head separately from the 19-channel weather and coordinate sequence.

Network parameters are optimized with AdamW at an initial learning rate of 3×10^{-4} and weight decay of 10^{-4} (Géron, 2025). Gradient norms are clipped at one. Training runs for at most 40 epochs. A plateau scheduler halves the learning rate after two epochs without improvement, down to 10^{-6}. Early stopping retains the best weights after five epochs without improvement. For each test year, the preceding three historical years

provide the inner stopping partition; older observations train the network. Table 4.1 summarizes the forecasting suite.

Table 4.1. Comparative architectural summary of candidate forecasting models.

Model	Family	Input Space	Regularization	Tuning Parameters	Agronomic Function
$\mathcal{B}_{\text{naive}}$	Secular Baseline	None ($P_{\text{met}} = 0$)	Deterministic	None	Zero-weather trend baseline ($R^2_{\text{OOS}} = 0$).
ElasticNet	Regularized Linear	Tabular 2D (P_H)	$L_1 + L_2$ convex loss	λ, α	Linear baseline with collinearity grouping.
Random Forest	Bagging Trees	Tabular 2D (P_H)	p_{split}, leaf size	p_{split}, n_{leaf}, $d_{\max}$	Non-parametric partition; variance reduction.
XGBoost	Gradient Boosting	Tabular 2D (P_H)	λ_{reg}, γ_{split}, shrinkage	η, $d_{\max}$, colsample, λ, α_{reg}	Additive boosting; non-linear interaction.
LSTM (Linear)	Recurrent Deep	Sequence 3D (10d)	Dropout, weight decay	d_h, l, dr, bs	Decad sequential tracking; linear projection for integrated persistence.
LSTM (PReLU)	Recurrent Deep	Sequence 3D (10d)	Dropout, weight decay	d_h, l, dr, bs	Recurrent weather-only specification with a PReLU prediction head.

Source: Author's synthesis of candidate forecasting methodologies.

4.4 Chronological Expanding-Window Validation Protocol

Rigorous out-of-sample evaluation requires preventing information leakage across contiguous counties and chronological periods. We structure validation through an expanding-window protocol.

4.4.1 Three-Phase Temporal Splitting Scheme

Random observation splits leak spatial weather signals across contiguous counties (Sweet et al., 2023). We enforce a three-phase chronological expanding-window partition (Table 4.2):

1. **Phase 1: Historical Baseline Training (1951–1984):** 34 years (4,590 observations) establishing base climatology, secular trends, and initial model weights.

2. **Phase 2: Hyperparameter Validation (1985–1995):** Phase 2 (1985–1995) deliberately incorporates three of the most severe macro-climatic shocks in modern agricultural history: the historic 1988 drought (−25.6%), the 1993 Mississippi River flood (−10.5%), and the 1994 record bumper harvest (+10.7%). Exposing candidate

architectures to wide empirical variance during hyperparameter tuning prevents algorithms from over-regularizing and collapsing predictions toward historical county means (conditional shrinkage), ensuring that the complexity ladder is calibrated to preserve sensitivity to acute downside shocks before entering the modern 2011–2025 test window.

3. **Phase 3: Out-of-Sample Expanding-Window Testing (2011–2025):** 15 modern harvest campaigns, providing 2,025 county-year evaluations per horizon. Training expands across $\{1951, \ldots, Y-1\}$ prior to each test year, incorporating historical data sequentially. This evaluates held-out years strictly out-of-sample across contemporary drought extremes (2012, 2023), planting disruptions (2019), and record bumper harvests (2016, 2021).

Table 4.2. Three-phase chronological validation and expanding-window testing framework.

Phase / Fold	Training Partition Span	Target Evaluation Span
Phase 1 (Base)	1951–1984 (34 years, 4,590 county-year obs)	Baseline historical model fitting.
Phase 2 (Val)	1951–1984 training pool	1985–1995 (11 years, 1,485 obs; validation and tuning).
Fold 1 (2011)	1951–2010 (60 years, 8,100 county-year obs)	2011 out-of-sample test (135 county evaluations).
Fold 2 (2012)	1951–2011 (61 years, 8,235 county-year obs)	2012 out-of-sample test (135 obs, Flash Drought).
	.....	
Fold 15 (2025)	1951–2024 (74 years, 9,990 county-year obs)	2025 out-of-sample test (135 county evaluations).

Source: Author's validation architecture.

4.4.2 Strict Train-Only Preprocessing Isolation

We compute all transformations strictly within each training fold to eliminate look-ahead bias:

- Secular trend parameters $(\hat{\alpha}_c^{(Y)}, \hat{\beta}_c^{(Y)})$ are estimated exclusively on historical years $\{1951, \ldots, Y-1\}$.
- Weather features are standardized using training partition moments:

$$\tilde{w}_{k,j} = \frac{w_{k,j} - \mu_{k,j}^{\text{train}}}{\sigma_{k,j}^{\text{train}}} \tag{4.29}$$

Moments μ^{train} and σ^{train} apply to test year Y without updating.

4.4.3 Hyperparameter Tuning via Grid Search

We conduct hyperparameter tuning across validation partition $\mathcal{T}_{\text{val}}$ (1985–1995, 1,485 observations) using chronological expanding-window grid search. For each candidate configuration and validation campaign $Y_{\text{val}} \in \{1985, \ldots, 1995\}$, models train on historical campaigns $\{1951, \ldots, Y_{\text{val}} - 1\}$ with train-only detrending and scaling, predicting year Y_{val}. Table 4.3 summarizes the search spaces.

Table 4.3. Pre-specified hyperparameter search spaces for expanding-window validation grid search.

Model	Hyperparameter	Search Space	Combinations
ElasticNet	Regularization (λ) Penalty Ratio (α)	$\{10^{-3}, 10^{-2}, 10^{-1}, 1.0, 5.0\}$ $\{0.1, 0.3, 0.5, 0.7, 0.9\}$	25
Random Forest	Split Features (p_{split}) Leaf Samples (n_{leaf}) Maximum Depth ($d_{\max}$)	{'sqrt', 0.33, 0.5} $\times P_H$ {5, 20} {8, 14, None}	18
XGBoost	Learning Rate (η) Maximum Depth ($d_{\max}$) L_2 Leaf Weight (λ_{reg}) L_1 Leaf Penalty (α_{reg})	{0.03, 0.06} {3, 5} {1.0, 5.0} {0.0, 0.5}	16
LSTM	Hidden Dimension (d_h) Recurrent Layers (l) Dropout Rate (dr) Batch Size (bs)	{64, 128, 256} {1, 2} {0.0, 0.2, 0.4} {25, 64}	44

Source: Author's specification following domain standards (Torsoni et al., 2023; Sharma et al., 2025; Géron, 2025). In Random Forest, $B = 100$ trees are used during validation tuning and $B = 300$ during final test evaluation. In XGBoost, feature sampling is fixed at $0.7 \times P_H$, row subsampling at 0.8, and leaf weight threshold at min_child_weight $= 3$. In LSTM, learning rate is self-calibrated via dynamic plateau scheduling and early stopping on inner validation folds.

The optimal parameter vector $\boldsymbol{\Theta}_H^*$ for horizon H minimizes pooled validation RMSE across the 11 expanding validation folds:

$$\boldsymbol{\Theta}_H^* = \arg\min_{\boldsymbol{\Theta} \in \mathcal{G}} \sqrt{\frac{1}{C \cdot |\mathcal{T}_{\text{val}}|} \sum_{Y_{\text{val}}=1985}^{1995} \sum_{c=1}^{C} \left(\hat{\epsilon}_{c,Y_{\text{val}}}^{(H)}(\boldsymbol{\Theta}) - \epsilon_{c,Y_{\text{val}}} \right)^2} \tag{4.30}$$

Selected hyperparameters remain fixed during the subsequent out-of-sample expanding-window evaluation across 2011–2025, where models re-estimate weights over $\{1951, \ldots, Y-1\}$ prior to predicting test year Y.

4.5 Performance Evaluation and Tail-Risk Testing

To evaluate forecasting skill and stress-test model resilience under acute climatic stress, we formalize continuous loss metrics, tail-risk benchmarks, and spatial inferential procedures.

4.5.1 Continuous Loss Metrics and Aggregation Hierarchy

Out-of-sample performance is evaluated on continuous yield anomaly prediction error:

$$e_{c,Y}^{(H)} = \hat{\epsilon}_{c,Y}^{(H)} - \epsilon_{c,Y} = \hat{y}_{c,Y}^{(H)} - y_{c,Y} \tag{4.31}$$

Because trend extrapolation $\hat{\tau}_{c,Y}$ is identical across models, evaluating errors on $\epsilon_{c,Y}$ is mathematically equivalent to evaluating errors on total yield levels $y_{c,Y}$, without secular trend variance inflating reported accuracy.

Predictions are generated for each individual county c and year Y. To synthesize performance across the panel, continuous loss metrics are evaluated over all $N_{\text{test}} = 15$ test years ($M_{\text{test}} = 2{,}025$ county-year evaluations per lead time):

$$\text{RMSE}_{\text{OOS}}(H) = \sqrt{\frac{1}{M_{\text{test}}} \sum_{Y=2011}^{2025} \sum_{c=1}^{C} \left(\hat{\epsilon}_{c,Y}^{(H)} - \epsilon_{c,Y} \right)^2} \tag{4.32}$$

$$\text{MAE}_{\text{OOS}}(H) = \frac{1}{M_{\text{test}}} \sum_{Y=2011}^{2025} \sum_{c=1}^{C} \left| \hat{\epsilon}_{c,Y}^{(H)} - \epsilon_{c,Y} \right| \tag{4.33}$$

$$R^2_{\text{OOS}}(H) = 1 - \frac{\sum_{Y=2011}^{2025} \sum_{c=1}^{C} \left(\epsilon_{c,Y} - \hat{\epsilon}_{c,Y}^{(H)} \right)^2}{\sum_{Y=2011}^{2025} \sum_{c=1}^{C} \epsilon_{c,Y}^2} \tag{4.34}$$

$$\text{Skill}_{\text{RMSE}}(H) = 1 - \frac{\text{RMSE}_{\text{model}}(H)}{\text{RMSE}_{\text{naive}}} \tag{4.35}$$

We designate RMSE_{OOS} as the primary loss metric because quadratic loss heavily penalizes large forecast errors during acute yield shock regimes, while MAE_{OOS} provides a robust baseline metric of typical absolute dispersion.

To assess regional stability without presenting 135 separate county tables, we compute cross-sectional annual averages:

$$\overline{\text{RMSE}}_Y(H) = \sqrt{\frac{1}{C} \sum_{c=1}^{C} \left(\hat{\epsilon}_{c,Y}^{(H)} - \epsilon_{c,Y} \right)^2}, \quad \text{Bias}_Y(H) = \frac{1}{C} \sum_{c=1}^{C} \left(\hat{\epsilon}_{c,Y}^{(H)} - \epsilon_{c,Y} \right) \tag{4.36}$$

tracking how predictive accuracy and directional bias evolve during extreme shortfall and bumper seasons.

4.5.2 Out-of-Sample Tail-Risk Stress Testing

To evaluate model resilience during extreme disasters without arbitrary discretization bins, predictions are stress-tested across benchmark historical shock cohorts in the 2011–2025 test window:

- **Adverse Shock Cohorts (Shortfall Campaigns):** The 2012 midsummer flash

drought (−11.7% regional anomaly), the 2019 widespread planting crisis, and the 2023 late-summer flash drought.

- **Favorable Shock Cohorts (Bumper Campaigns):** The modern bumper harvests of 2016 (+10.8%) and 2021 (+10.8%).

This protocol evaluates whether flexible regression estimators capture physical shock amplitudes or exhibit conditional shrinkage toward the historical mean.

4.5.3 Statistical Inference: Pairwise Diebold–Mariano Tests

To assess whether predictive gains over the naive baseline are statistically significant and distinguishable from sample noise, we evaluate Diebold and Mariano (1995) test statistics on cross-sectionally averaged annual loss differentials:

$$d_Y(H) = \frac{1}{C}\sum_{c=1}^{C}\left[\left(e_{c,Y}^{\text{baseline}}(H)\right)^2 - \left(e_{c,Y}^{\text{model}}(H)\right)^2\right] \tag{4.37}$$

Defining $\bar{d}(H) = \frac{1}{N_{\text{test}}}\sum_{Y=2011}^{2025} d_Y(H)$ as the mean differential across the $N_{\text{test}} = 15$ evaluation years, the test statistic is:

$$S_{\text{DM}}(H) = \frac{\bar{d}(H)}{\sqrt{\widehat{\text{Var}}(\bar{d}(H))}} \tag{4.38}$$

where the variance of the mean differential is estimated using a Newey–West (HAC) covariance estimator to account for potential serial persistence across consecutive campaigns. We report one-sided p-values testing the null hypothesis of equal forecast accuracy against the alternative of superior accuracy by the weather-augmented model.

4.5.4 Residual Diagnostics

Valid statistical inference requires verifying that out-of-sample forecast residuals $e_{c,Y}^{(H)}$ do not exhibit persistent temporal autocorrelation or uncontrolled spatial clustering:

1. **Temporal Autocorrelation (AR(1)):** For each individual county c, we compute the first-order serial correlation coefficient of out-of-sample forecast errors across the 15 test years:

$$\widehat{\rho}_{c,1}^{(H)} = \frac{\sum_{Y=2012}^{2025}(e_{c,Y}^{(H)} - \bar{e}_c^{(H)})(e_{c,Y-1}^{(H)} - \bar{e}_c^{(H)})}{\sum_{Y=2011}^{2025}(e_{c,Y}^{(H)} - \bar{e}_c^{(H)})^2} \tag{4.39}$$

   The panel-level temporal persistence $\overline{\text{AR}(1)}^{(H)} = \frac{1}{C}\sum_{c=1}^{C}\widehat{\rho}_{c,1}^{(H)}$ tests whether multi-year agronomic memory or rotational inertia remains uncaptured by the feature space.

2. **Spatial Autocorrelation (Moran's I):** To measure cross-sectional spatial clustering of forecast errors across contiguous counties, we compute Moran's I for each test year Y:

$$I_Y^{(H)} = \frac{C}{\sum_{i=1}^{C}\sum_{j=1}^{C} w_{ij}} \frac{\sum_{i=1}^{C}\sum_{j=1}^{C} w_{ij}(e_{i,Y}^{(H)} - \bar{e}_Y^{(H)})(e_{j,Y}^{(H)} - \bar{e}_Y^{(H)})}{\sum_{i=1}^{C}(e_{i,Y}^{(H)} - \bar{e}_Y^{(H)})^2} \tag{4.40}$$

where $w_{ij} = 1/d_{ij}$ $(i \neq j,\ w_{ii} = 0)$ represents the row-standardized inverse Euclidean distance between county centroids $(\tilde{\text{Lat}}_c, \tilde{\text{Lon}}_c)$ (Sweet et al., 2023). Values of Moran's $I \in [-1, 1]$ near zero indicate spatial randomness of residuals, whereas positive values indicate geographic clustering of errors (suggesting that correlated local shocks, microclimates, or unmodelled spatial heterogeneity remain in the error covariance structure). Annual estimates are averaged across the test window: $\overline{I}^{(H)} = \frac{1}{N_{\text{test}}} \sum_{Y=2011}^{2025} I_Y^{(H)}$.

5. Empirical Results

Evaluations are calculated on the expanding test partition spanning harvest years 2011–2025 ($N_{\text{test}} = 15$ campaigns, $M_{\text{test}} = 2{,}025$ county-year evaluations per lead time), with model hyperparameters fixed on the 1985–1995 validation partition across all twelve forecast horizons ($H = 12 \dots 1$).

5.1 Lead-Time Dynamics and Progression across Horizons

At $H = 12$ forecasts predict a zero yield anomaly ($\hat{\epsilon}_{c,Y} = 0$) by construction, projecting solely the county-specific linear technological trend $\hat{\tau}_{c,Y}$. The out-of-sample root mean squared error of this naive benchmark defines the baseline error reference across the 2011–2025 test panel:

$$\text{RMSE}_{\text{naive}} = \sqrt{\frac{1}{M_{\text{test}}} \sum_{Y=2011}^{2025} \sum_{c=1}^{C} \epsilon_{c,Y}^2} = 6.12 \text{ bu/ac}, \quad R^2_{\text{OOS}}(H = 12) \equiv 0.00 \tag{5.1}$$

This baseline provides the normalizing denominator in Equation (4.34). Positive R^2_{OOS} values at subsequent horizons ($H = 11 \dots 1$) consequently quantify the incremental predictive skill contributed by observed agrometeorological conditions beyond the trend baseline.

Table 5.1 compares candidate models across the twelve forecast horizons under two information sets: a pure meteorological feature space and an integrated specification incorporating the prior-year detrended yield anomaly ($\epsilon_{c,Y-1}$). At the final horizon ($H = 1$), all specifications implement harvest closure $H^*(s)$, freezing the information set on September 30 for Minnesota (Maturity Group I) and October 31 for the remaining states to eliminate post-harvest atmospheric noise. Consistent with the validation tuning in Section 4.3.4, the weather-only LSTM employs a PReLU projection head, while the integrated architecture uses a linear projection head.

The empirical performance matrix in Table 5.1 and corresponding trajectories in Figure 5.1 track predictive skill across the production cycle at county level ($M_{\text{test}} = 2{,}025$ out-of-sample county-year evaluations). County-level evaluation measures local predictive accuracy directly, without the variance reduction that occurs when averaging across counties. Across the twelve monthly horizons, performance follows three distinct chronological regimes:

Before May ($H = 11 \dots 7$, Dec–Apr), weather-only models do not improve upon the naive baseline. These monthly forecast origins precede the primary planting window documented by Hoffman et al. (2020). Integrated ElasticNet remains close to the baseline ($R^2_{\text{OOS}} \approx 0.00$), while flexible non-linear estimators generate negative scores, reflecting

Table 5.1. Pooled county-level out-of-sample R^2 across forecast horizons and information sets (2011–2025).

		Weather-Only (R^2_{OOS})					Integrated (+Lag, R^2_{OOS})				
H	**Cutoff Date**	**ENet**	**RF**	**XGB**	**LSTM**	**Ens**	**ENet**	**RF**	**XGB**	**LSTM**	**Ens**
Pre-Sowing Horizons ($H = 11 \ldots 7$)											
$H = 11$	Dec 31	-0.02	-0.05	-0.19	-0.15	-0.07	**+0.00**	-0.04	-0.14	-0.18	-0.07
$H = 10$	Jan 31	**-0.01**	-0.10	-0.10	-0.22	-0.10	**-0.01**	-0.09	-0.07	-0.21	-0.09
$H = 9$	Feb 28/29	-0.04	-0.11	-0.09	-0.12	-0.06	**-0.03**	-0.10	-0.16	-0.15	-0.07
$H = 8$	Mar 31	-0.02	-0.04	-0.05	-0.20	-0.09	**-0.00**	-0.05	-0.04	-0.19	-0.07
$H = 7$	Apr 30	-0.02	-0.07	-0.05	-0.22	-0.09	**-0.00**	-0.05	-0.03	-0.17	-0.06
Crop Campaign ($H = 6 \ldots 1$)											
$H = 6$	May 31	+0.00	+0.00	+0.01	-0.05	+0.00	+0.00	+0.03	**+0.05**	+0.00	+0.03
$H = 5$	Jun 30	+0.11	+0.07	+0.06	+0.07	+0.11	+0.13	+0.07	+0.08	+0.14	**+0.16**
$H = 4$	Jul 31	+0.21	+0.12	+0.16	+0.15	+0.21	+0.22	+0.15	+0.17	+0.23	**+0.25**
$H = 3$	Aug 31	+0.28	+0.21	+0.25	+0.30	+0.31	+0.30	+0.22	+0.28	+0.29	**+0.32**
$H = 2$	Sep 30	+0.30	+0.23	+0.27	+0.33	+0.35	+0.32	+0.23	+0.27	+0.33	**+0.36**
$H = 1$	$H^*(s)$	+0.29	+0.21	+0.27	+0.34	+0.35	+0.32	+0.24	+0.30	+0.37	**+0.38**

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data, 2011–2025 out-of-sample forecasts. Weather-only specifies ERA5-Land predictors; Integrated specifications add lagged anomaly ϵ_{t-1}.

variance inflation and overfitting on uninformative antecedent noise.

With the inclusion of May planting-window observations ($H = 6$, May 31), initial predictive skill begins to emerge across candidate architectures: pure meteorological XGBoost crosses into positive territory at $R^2_{\text{OOS}} = +0.01$ (rising to +0.05 with prior-year carryover). This marks the operational inflection point where environmental observations first surpass the naive baseline.

Predictive accuracy accelerates as reproductive weather accumulates across summer ($H = 5 \ldots 1$, Jun–Oct). Substantial performance gains concentrate during flowering (July, $H = 4$) and pod filling (August, $H = 3$), when moisture deficits and thermal stress directly dictate seed mass. Across all candidate architectures, skill reaches a plateau proximate to physiological maturity, with marginal increments between September and October.

Incorporating prior-year detrended yield memory (ϵ_{t-1}) provides a consistent, incremental improvement of approximately +0.03 points in R^2_{OOS} across all model families at harvest closure (ElasticNet rises from 0.29 to 0.32, LSTM from 0.34 to 0.37, and the ensemble from 0.35 to 0.38). In contrast to previous studies that report massive variance gains from autoregressive lags on raw yields (Chen and Zhang, 2026), our controlled detrending isolates genuine multi-year agronomic memory—such as deep soil moisture carryover and crop rotational inertia—from historical technological drift.

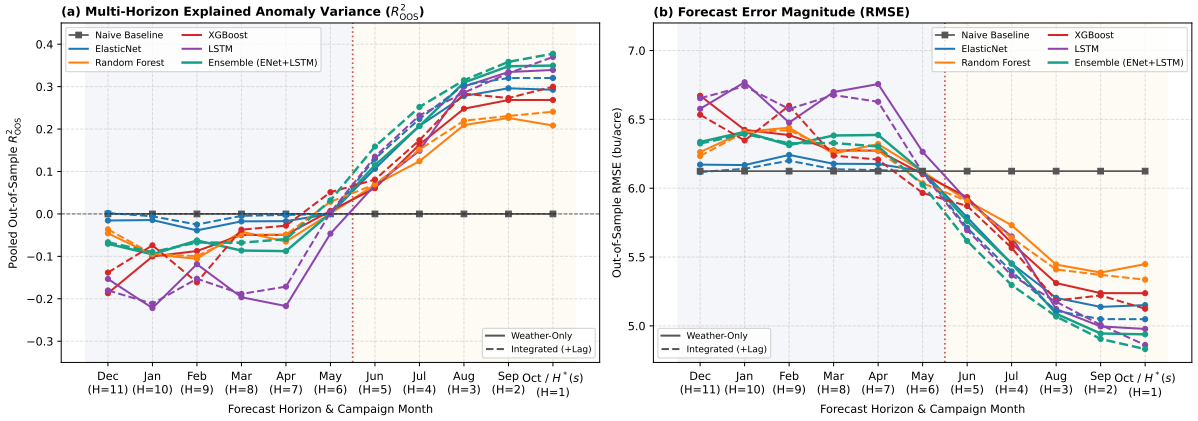


Figure 5.1. Progression of out-of-sample predictive performance across monthly lead times ($H = 11 \ldots 1$) for weather-only (solid lines) and lag-integrated (dashed lines) specifications at county level (2011–2025): (a) R^2_{OOS}; (b) RMSE (bu/ac).

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data, 2011–2025.

5.2 Comparative Model Performance across Active Horizons

Across the active growing season ($H = 5 \ldots 1$, June–October), out-of-sample forecast accuracy reveals three core empirical insights on model complexity and inductive biases:

First, penalized linear regression (ElasticNet) achieves the highest individual accuracy during vegetative growth and flowering ($H = 5 \ldots 3$, June to August), reaching $R^2_{\text{OOS}} = 0.30$ (0.28 weather-only). When the physical signal is only partially accumulated, regularized shrinkage restricts estimation variance on high-dimensional climate inputs (James et al., 2023). Non-linear tree ensembles (Random Forest and XGBoost) do not improve upon ElasticNet on monthly tabular features, yielding lower out-of-sample scores ($R^2_{\text{OOS}} \leq 0.28$).

Second, the recurrent neural network (LSTM) accelerates skill as reproductive weather accumulates, matching ElasticNet in August ($H = 3, R^2_{\text{OOS}} = 0.29$–$0.30$) and surpassing it from September onward ($H = 2$, reaching $R^2_{\text{OOS}} = 0.33$, and $H = 1$, reaching 0.37 with lag vs 0.34 weather-only). While monthly tabular models aggregate daily weather into 30-day blocks, the LSTM processes 10-day decad sequences, capturing the timing and duration of temperature and moisture stress during pod filling without increasing feature dimensionality.

Third, the equal-weighted convex ensemble (50% ElasticNet + 50% LSTM) consistently achieves champion performance across active horizons, culminating in $R^2_{\text{OOS}} = 0.38$ at harvest closure (0.35 under weather-only forcing). This synergy stems from the empirical orthogonality of model forecast residuals ($r < 0.70$), combining the variance stability of linear shrinkage with the sequence sensitivity of recurrent networks.

Having established the empirical gain of prior-year detrended yield carryover (ϵ_{t-1}), all subsequent stress tests, macro-regional evaluations, and diagnostic attributions focus on

the lag-integrated model specifications (referred to simply as ElasticNet, Random Forest, XGBoost, LSTM, and Ensemble).

5.3 Inferential Significance and Tail-Risk Stress Testing

5.3.1 Pairwise Predictive Superiority: Diebold–Mariano Tests

To establish whether predictive gains over the naive baseline and between competing model families are statistically distinguishable from sample variability, we evaluate one-sided Diebold and Mariano (1995) tests on annual mean squared loss differentials across the 15 out-of-sample test years (2011–2025). Following the protocol formalized in Section 4.5.3, squared loss differentials are first averaged across the 135 counties within each crop year to account for contemporaneous spatial correlation, and test statistics are estimated with Newey–West HAC standard errors.

At the June emergence horizon ($H = 5$), weather-only ElasticNet and integrated LSTM achieve statistically significant loss reductions over the naive trend benchmark, with nominal p-values of 0.0011 and 0.00025 (DM = 3.06 and 3.48). Conversely, tree ensembles fail to reject the null hypothesis of equal accuracy at the 5% level (Random Forest $p = 0.074$; XGBoost $p = 0.094$). From July ($H = 4$) onward, all evaluated weather-informed models reject equal accuracy against the naive baseline ($p < 0.0001$).

At harvest closure ($H = 1$), pairwise testing between LSTM and ElasticNet yields DM = 0.99 ($p = 0.162$). Despite a +0.050 point improvement in R^2_{OOS} and a 0.188 bu/ac reduction in RMSE, the sequential deep architecture does not achieve statistically significant superiority over penalized linear shrinkage under serial correlation controls. The Linear-Recurrent Ensemble significantly improves upon ElasticNet at the 5% level (DM = 2.11, $p = 0.017$), while its performance difference from LSTM remains statistically indistinguishable (DM = 0.33, $p = 0.371$). These tests confirm that while meteorological forcing provides unequivocal predictive skill, pooled point rankings between flexible neural networks and regularized linear baselines do not establish uniform statistical dominance.

5.3.2 County-Level Decile Calibration and Residual Diagnostics

Table 5.2 reports mean predictions within deciles of realized anomalies at the June, August, and harvest-closure cutoffs, assessing tail calibration across the full test panel.

The decile breakdown in Table 5.2 and diagnostic suite in Figure 5.2 evaluate how effectively meteorological observations calibrate predictions across the yield distribution. Grouping predictions by realized anomaly deciles reveals a structural trade-off between conditional variance reduction and tail-risk sensitivity under squared-error loss.

Across distribution extremes, model predictions exhibit systematic conditional shrinkage. For both severe downside shocks (D_1, realized mean −8.82 bu/ac) and bumper harvests (D_{10}, realized mean +11.01 bu/ac), predictions consistently compress toward the zero-anomaly baseline. At harvest closure, LSTM and the ensemble predict −2.18

Table 5.2. Mean forecast anomalies (bu/ac) within realized anomaly deciles at three cutoffs (2011–2025).

Decile	Range (bu/ac)	Mean Obs.	ENet	RF	XGB	LSTM	Ens
Panel A: Early Vegetative Horizon ($H = 5$, June 30)							
D_1 (Severe Shortfall)	$[-22.85, -5.22]$	-8.82	-1.69	-1.78	**-2.08**	-0.92	-1.30
D_2	$[-5.15, -1.95]$	-3.43	-0.67	-0.99	**-1.14**	-0.69	-0.68
D_3	$[-1.92, +0.02]$	-0.93	-0.26	**-0.57**	-0.46	-0.09	-0.18
D_4	$[+0.02, +1.71]$	$+0.92$	$+0.13$	-0.23	$+0.03$	$+$**0.29**	$+0.21$
D_5	$[+1.72, +3.15]$	$+2.48$	$+0.26$	-0.14	$+0.04$	$+$**0.54**	$+0.40$
D_6	$[+3.16, +4.40]$	$+3.80$	$+0.55$	$+0.20$	$+0.54$	$+$**0.88**	$+0.72$
D_7	$[+4.42, +5.59]$	$+5.02$	$+0.67$	$+0.32$	$+$**0.72**	$+$**0.72**	$+0.70$
D_8	$[+5.60, +6.83]$	$+6.29$	$+0.59$	$+0.32$	$+0.60$	$+$**0.88**	$+0.73$
D_9	$[+6.84, +8.83]$	$+7.77$	$+1.01$	$+0.60$	$+0.84$	$+$**1.33**	$+1.17$
D_{10} (Peak Yield)	$[+8.83, +22.93]$	$+11.01$	$+0.77$	$+0.18$	$+0.37$	$+$**0.90**	$+0.84$
Panel B: Pod-Filling Horizon ($H = 3$, August 31)							
D_1 (Severe Shortfall)	$[-22.85, -5.22]$	-8.82	-2.19	-1.37	-1.77	**-2.61**	-2.40
D_2	$[-5.15, -1.95]$	-3.43	-0.84	-0.59	-0.59	**-1.32**	-1.08
D_3	$[-1.92, +0.02]$	-0.93	-0.14	-0.12	-0.15	**-0.61**	-0.38
D_4	$[+0.02, +1.71]$	$+0.92$	$+0.56$	$+0.34$	$+$**0.73**	$+0.26$	$+0.41$
D_5	$[+1.72, +3.15]$	$+2.48$	$+0.71$	$+0.39$	$+$**0.73**	$+0.55$	$+0.63$
D_6	$[+3.16, +4.40]$	$+3.80$	$+1.11$	$+0.68$	$+$**1.23**	$+1.13$	$+1.12$
D_7	$[+4.42, +5.59]$	$+5.02$	$+1.46$	$+0.97$	$+$**1.66**	$+1.42$	$+1.44$
D_8	$[+5.60, +6.83]$	$+6.29$	$+1.56$	$+1.01$	$+$**1.72**	$+1.38$	$+1.47$
D_9	$[+6.84, +8.83]$	$+7.77$	$+1.78$	$+1.23$	$+$**1.84**	$+1.65$	$+1.72$
D_{10} (Peak Yield)	$[+8.83, +22.93]$	$+11.01$	$+$**2.33**	$+1.29$	$+2.11$	$+1.84$	$+2.09$
Panel C: Harvest Closure Horizon ($H^*(s)$)							
D_1 (Severe Shortfall)	$[-22.85, -5.22]$	-8.82	-2.13	-1.49	-1.59	**-2.18**	-2.16
D_2	$[-5.15, -1.95]$	-3.43	**-0.83**	-0.54	-0.43	-0.60	-0.72
D_3	$[-1.92, +0.02]$	-0.93	**-0.24**	-0.18	-0.19	-0.13	-0.18
D_4	$[+0.02, +1.71]$	$+0.92$	$+0.70$	$+0.43$	$+0.60$	$+$**0.85**	$+0.78$
D_5	$[+1.72, +3.15]$	$+2.48$	$+1.05$	$+0.51$	$+0.84$	$+$**1.21**	$+1.13$
D_6	$[+3.16, +4.40]$	$+3.80$	$+1.50$	$+0.69$	$+1.29$	$+$**1.69**	$+1.60$
D_7	$[+4.42, +5.59]$	$+5.02$	$+1.84$	$+1.14$	$+1.86$	$+$**2.07**	$+1.96$
D_8	$[+5.60, +6.83]$	$+6.29$	$+1.98$	$+1.27$	$+1.79$	$+$**2.34**	$+2.16$
D_9	$[+6.84, +8.83]$	$+7.77$	$+2.21$	$+1.36$	$+2.01$	$+$**2.55**	$+2.38$
D_{10} (Peak Yield)	$[+8.83, +22.93]$	$+11.01$	$+$**2.86**	$+1.61$	$+2.33$	$+2.84$	$+2.85$

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data, 2011–2025 out-of-sample forecasts. Ens: ensemble.

and -2.16 bu/ac in D_1, and $+2.85$ bu/ac in D_{10}. This attenuation reflects standard L_2 conditional-mean shrinkage: because extreme agricultural shocks involve unobserved local factors such as subsurface drainage, seed genetics, and localized pest pressure, estimators trained under mean squared error naturally shrink predictions toward zero to avoid excessive variance penalties. The models correctly anticipate the physical sign and direction of the stress, substantially reducing quadratic loss relative to the naive benchmark.

Conversely, across intermediate regimes (D_2–D_9), predicted values scale monotonically with realized anomalies as reproductive weather accumulates between June ($H = 5$) and harvest ($H = 1$). Residual Q-Q plots in Figure 5.2(b) confirm that forecast departures remain symmetric around zero, with homoskedastic error dispersion across fitted values (panel c).

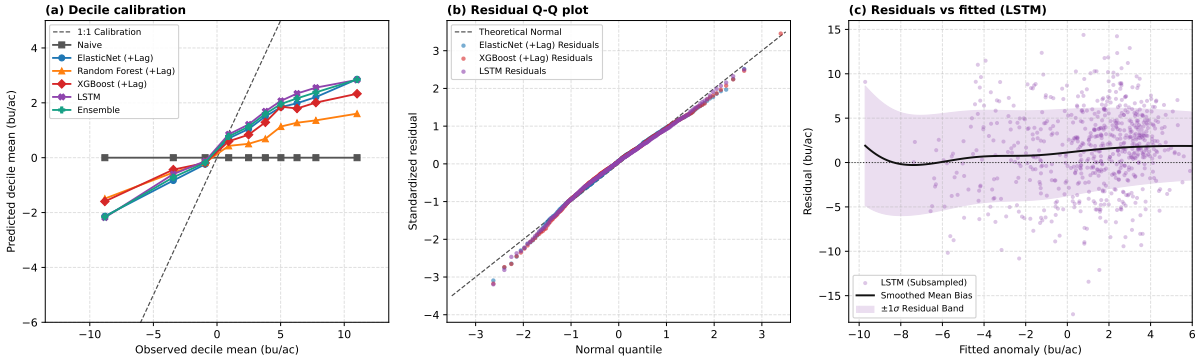


Figure 5.2. Residual and calibration diagnostics at harvest closure horizon $H^*(s)$: decile calibration across anomaly quantiles ($D_1 \dots D_{10}$), residual Q-Q distribution, and residual variance across fitted values.

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data, out-of-sample test evaluations.

Beyond tail calibration, evaluating the spatiotemporal independence of out-of-sample residuals validates model specification. At harvest closure, mean county AR(1) coefficients remain close to zero (-0.054 for ElasticNet, -0.062 for XGBoost, and -0.058 for LSTM). These near-zero values confirm that detrending and one-year anomaly lags absorb interannual memory, leaving no unresolved autoregressive persistence across the 15-year test window.

Concurrently, spatial diagnostic tests reveal residual geographic clustering. In the naive baseline, forecast errors exhibit significant spatial clustering with a mean Moran coefficient of 0.183. Conditioning on ERA5-Land weather features absorbs large-scale atmospheric patterns, reducing Moran's I to 0.142 in ElasticNet and 0.145 in LSTM (0.171 in XGBoost). However, the index does not collapse to zero: a moderate positive spatial correlation persists. This residual clustering demonstrates that while gridded reanalysis captures macro-regional weather fronts, contiguous counties share unmodelled spatial determinants—such as contiguous soil associations, regional aquifer access, and common agricultural management practices—that induce correlated local errors.

5.3.3 Historical Shock Regimes: 2012 Drought versus 2016 Bumper Season

To evaluate how early in the growing season systemic shocks become detectable at county level, Table 5.3 breaks down out-of-sample forecast errors across active campaign months ($H = 5 \ldots 1$) for the two defining macro-climatic shock years of the test era: the 2012 flash drought (severe adverse shock, panel anomaly -11.7%, mean realized anomaly -4.07 bu/ac) and the 2016 bumper harvest (severe favorable shock, expanding-window test anomaly $+14.8\%$ relative to historical trend, mean realized anomaly $+7.34$ bu/ac).

Table 5.3. County-level forecast RMSE (bu/ac) during the 2012 drought and 2016 bumper season.

H	Cutoff Date	Naive Trend	ElasticNet	XGBoost	LSTM	Ens
Panel A: 2012 Flash Drought						
$H = 5$	Jun 30	7.20	**6.01**	6.08	6.53	6.16
$H = 4$	Jul 31	7.20	**5.80**	6.03	6.06	5.84
$H = 3$	Aug 31	7.20	5.72	5.81	**5.56**	5.58
$H = 2$	Sep 30	7.20	5.79	5.90	5.71	**5.51**
$H = 1$	$H^*(s)$	7.20	5.84	5.83	5.71	**5.66**
Panel B: 2016 Bumper Season						
$H = 5$	Jun 30	7.87	8.19	8.39	**7.57**	7.86
$H = 4$	Jul 31	7.87	**6.58**	6.61	7.30	6.89
$H = 3$	Aug 31	7.87	**5.13**	5.59	6.42	5.73
$H = 2$	Sep 30	7.87	**4.22**	6.15	5.32	4.71
$H = 1$	$H^*(s)$	7.87	**4.32**	5.17	4.48	4.33

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data, 2011–2025 out-of-sample forecasts. Ens: ensemble.

Tracking error trajectories across growing months in Table 5.3 reveals an informative empirical pattern across lead times. Across active campaign months, the lowest RMSE for champion specifications in these specific shock campaigns occurs at $H = 2$ (September 30). In the 2012 drought, the ensemble attains its lowest error of 5.51 bu/ac at $H = 2$, before recording 5.66 bu/ac at $H = 1$ (with ElasticNet moving from 5.79 to 5.84 bu/ac). In the 2016 bumper season, ElasticNet reaches an RMSE of 4.22 bu/ac at $H = 2$, compared to 4.32 bu/ac at $H = 1$.

Empirically, this localized leveling off between late September and late October reflects the underlying biophysical calendar of the crop. By the end of September ($H = 2$), physiological maturity (R7–R8) is largely completed across the major soybean-producing counties of the Corn Belt, meaning that subsequent October weather primarily governs field dry-down and harvest logistics without further biological yield accumulation. While the final harvest closure horizon $H = 1$ provides the complete full-campaign environmental record across all states, the predictive models have already assimilated the dominant environmental signals by early autumn, yielding stable performance with minor sample-

specific fluctuations across the final month.

In terms of lead-time shock detection, substantial error contraction occurs well ahead of harvest. In 2012, errors drop sharply between June ($H = 5, 6.16$ bu/ac) and August ($H = 3, 5.58$ bu/ac), confirming that the signature of the flash drought is already captured by late August. Similarly, in 2016, favorable conditions throughout July and August enable ElasticNet to contract its error from 8.19 bu/ac in June to 5.13 bu/ac in August and 4.22 bu/ac in September.

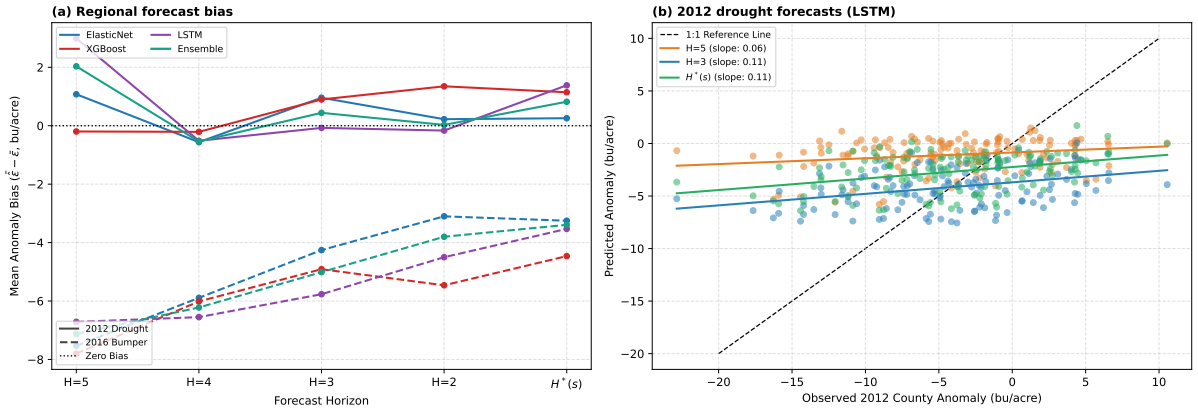


Figure 5.3. Lead-time bias convergence and county-level prediction resolution across active forecast horizons ($H = 5 \ldots 1$), contrasting regional mean anomaly bias trajectories with county-level scatter dynamics during the 2012 flash drought.

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data, out-of-sample test evaluations.

Figure 5.3(a) tracks regional bias convergence during the two shock campaigns, showing how directional bias contracts toward zero as reproductive weather unfolds. Panel (b) illustrates county prediction scatter during the 2012 drought: while predictions accurately track the downward shift, county-level scatter remains more compressed than realized departures. This compression reflects conditional-mean shrinkage under squared-error loss when localized microclimatic and agronomic drivers remain unobserved.

5.4 Agrometeorological Drivers and Attribution Across Scales

5.4.1 Attribution Methodology and Surrogate Recurrent Mapping

We evaluate agrometeorological sensitivities across model paradigms using Grouped Permutation Feature Importance (GPFI; Sweet et al., 2023) across 19 predictor groups: 17 weather indicators, lagged yield, and jointly permuted spatial coordinates on the 2011–2025 out-of-sample test panel. For each physical indicator j, all constituent monthly features across the full campaign ($m \in \{\text{Nov}, \ldots, \text{Oct}\}$) are permuted jointly across test instances (π_j). Grouping across all campaign months preserves the seasonal correlation profile while breaking the association with realized yield anomalies. Predictive degrada-

tion is quantified by out-of-sample explained variance loss:

$$\Delta R^2_{\mathrm{OOS}}(j) = \max\left(0,\, R^2_{\mathrm{OOS}}(\mathrm{base}) - R^2_{\mathrm{OOS}}(\pi_j)\right) \tag{5.2}$$

where the non-negativity constraint neutralizes sampling noise from uninformative features.

Because baseline predictive accuracy differs across candidate architectures, absolute variance losses are normalized into a Relative Information Share (%):

$$s_j = \frac{\Delta R^2_{\mathrm{OOS}}(j)}{\sum_{k=1}^{19} \Delta R^2_{\mathrm{OOS}}(k)} \times 100\% \tag{5.3}$$

Normalizing into relative information shares allows direct comparison of attentional distributions across model families.

For the sequential LSTM, permuting input channels across three-dimensional sequence tensors ($N \times 36 \times 19$) severely disrupts recurrent hidden state dynamics and entails prohibitive computational cost. We therefore implement a Global Surrogate Model (Sweet et al., 2023): a gradient-boosted tree ensemble (XGBoost) trained to replicate the predictions of the LSTM in anomaly space. The surrogate achieves near-perfect fidelity ($R^2 = 0.9968$ for $H = 1$ and $R^2 = 0.9998$ for $H = 2$, with RMSE < 0.18 bu/ac), legitimizing its use for post-hoc agrometeorological attribution.

Figure 5.4 reports the Relative Information Share across all 19 predictor groups for ElasticNet, XGBoost, and the LSTM surrogate at harvest closure horizon $H^*(s)$, while Figure 5.5 aggregates these individual shares into five coherent macro-agronomic domains.

Because atmospheric demand and hydrological supply variables are highly collinear, individual architectures select substitute proxies reflecting their inductive biases:

- **XGBoost** operates as an acute thermal-stress sensor: it is dominated by Atmospheric Demand and Heat (28.9%, driven by non-linear thresholds of VPD at 18.9%) and Thermal Regimes (34.6%), while assigning minimal weight to multi-year yield memory (4.1%).

- **ElasticNet** distributes predictive weights across Thermal Regimes (43.0%), Atmospheric Demand (25.6%), and Soil Moisture (20.9%), selecting dewpoint temperature (T_{dew}, 27.9%) as a parsimonious linear proxy for surface moisture.

- **LSTM** acts as a cumulative water balance integrator and memory anchor: predictive shares are led by Hydrological Supply (29.7%, dominated by $P - ET_0$ at 24.9%) and Memory and Spatial Baseline (25.1%, where lagged yield contributes 21.2%), capturing soil moisture recharge and multi-year agronomic persistence.

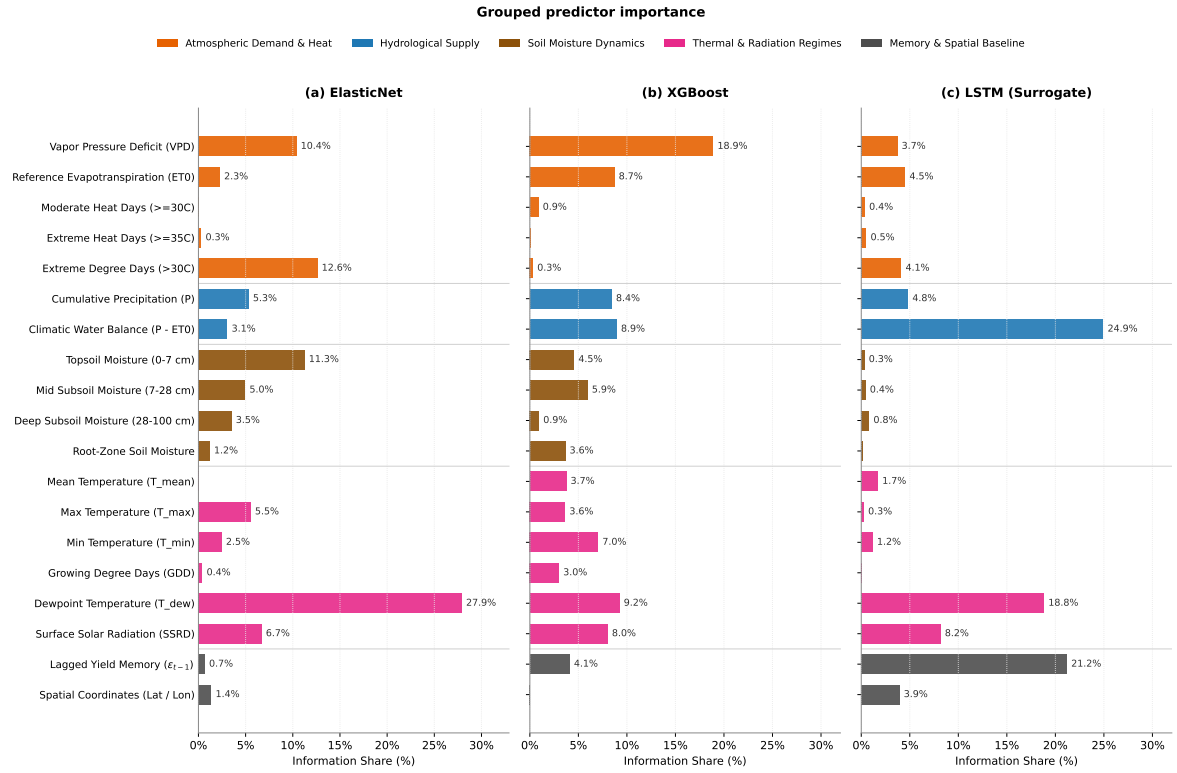


Figure 5.4. Grouped Permutation Feature Importance (GPFI) across 19 predictor groups at harvest closure $H^*(s)$ for benchmark models (2011–2025). Relative Information Share measures percentage contribution to total ΔR^2_{OOS} degradation.

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data, out-of-sample test evaluations 2011–2025.

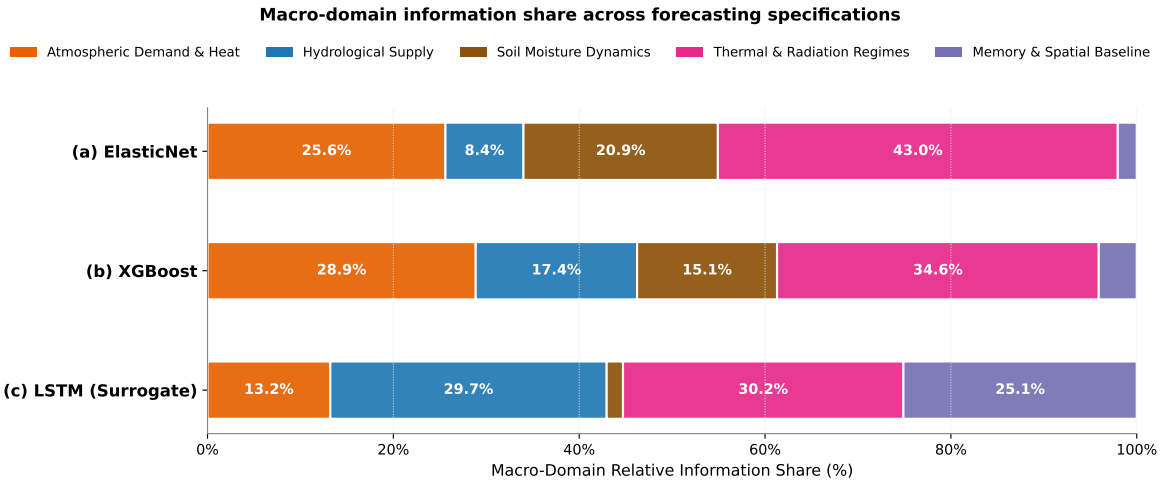


Figure 5.5. Relative information share by physical domain at harvest closure horizon $H^*(s)$ across benchmark architectures (2011–2025 test panel), evaluated via Grouped Permutation Feature Importance (GPFI).

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data, out-of-sample test evaluations 2011–2025.

5.4.2 Spatial Heterogeneity of Predictive Performance

Figure 5.6 maps county-level out-of-sample explained variance (R^2_{OOS}) for the sequential LSTM at harvest closure horizon $H^*(s)$ across all 135 study counties.

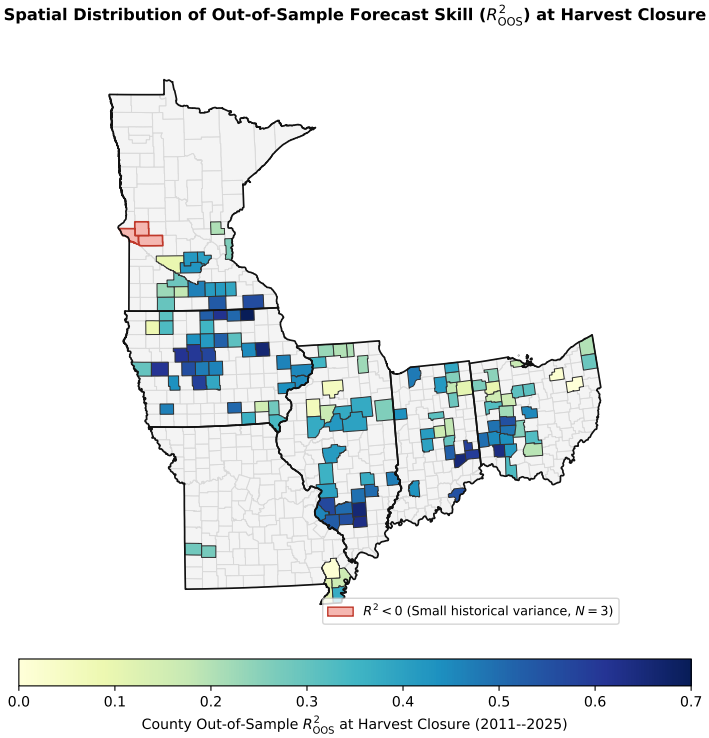


Figure 5.6. Spatial distribution of out-of-sample predictive performance (R^2_{OOS}) for the LSTM model at harvest closure horizon $H^*(s)$ across the 135 study counties (2011–2025).

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data, 2011–2025 out-of-sample test evaluations.

Out-of-sample performance demonstrates broad spatial robustness: 98% of the panel (132 out of 135 counties) achieves positive explained variance ($R^2_{\text{OOS}} > 0$) relative to the naive baseline. Across the study domain, the median county-level R^2_{OOS} is +0.386, with the 75th percentile reaching +0.475 and peak counties exceeding +0.70. State median scores are highest in Iowa (0.416), Indiana (0.412), and Illinois (0.337), where rainfed production aligns with regional climate forzings. The remaining three counties in western Minnesota record marginal negative scores due to near-zero historical anomaly variance, despite absolute errors (4.3 bu/ac) being fully in line with the panel average.

5.5 Anomaly Space versus Raw Yield Performance

We reconstruct physical yield forecasts by adding the training-only trend to the predicted anomaly, following Equation (4.3):

$$\hat{y}^{(H)}_{c,Y} = \hat{\tau}_{c,Y} + \hat{\epsilon}^{(H)}_{c,Y} \tag{5.4}$$

As established in Section 4.5.1, RMSE and MAE are algebraically identical between anomaly and reconstructed yield spaces because trend extrapolation $\hat{\tau}_{c,Y}$ is constant across models. Table 5.4 reports performance metrics across both spaces at harvest closure across the 2,025 test evaluations.

Table 5.4. Forecast accuracy in anomaly and reconstructed yield space at $H^*(s)$ (2011–2025).

Model	RMSE	MAE	R^2_{OOS}	R^2_{level}	r_{level}
Naive Secular Trend	6.124	5.050	0.000	0.474	0.746
ElasticNet	5.049	4.045	0.320	0.642	0.821
XGBoost	5.125	4.098	0.300	0.632	0.816
LSTM	4.861	3.847	0.370	0.668	0.833
Ensemble	4.831	3.861	0.378	0.673	0.838

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data, 2011–2025 out-of-sample forecasts. RMSE and MAE are in bu/ac.

Evaluating predictions on reconstructed physical yield levels ($\hat{y} = \hat{\tau} + \hat{\epsilon}$) clarifies the distinction between long-term agronomic progress and interannual weather skill (Table 5.4). The naive trend benchmark accounts for 47.4% of total raw yield variance ($r = 0.746$) across the test partition without ingesting meteorological data, capturing multidecadal genetic breeding, fertilizer efficiency, and permanent cross-county soil endowments.

Conditioning on ERA5-Land surface reanalysis and historical carryover raises explained variance to 66.8% ($r = 0.833$) for LSTM and 67.3% ($r = 0.838$) for the Linear-Recurrent Ensemble, resolving an incremental +19.9 percentage points of commercial yield variation beyond technological momentum.

Evaluating predictions on both levels also contextualizes comparisons with studies that benchmark models directly on raw physical yields (Chen and Zhang, 2026; Khaki et al., 2020). In raw yield space, high explained variance ($R^2 \geq 0.65$–0.75) reflects the joint contribution of multidecadal agronomic trends and interannual meteorological shocks. Projecting forecasts into anomaly space separates these two components, confirming that the empirical models capture genuine climate-induced departures from trend ($R^2_{\text{OOS}} = 0.378$) while preserving operational utility for physical production volume estimates.

5.6 Macro-Regional Aggregation and Supply-Chain Value

While county-level estimation ($M_{\text{test}} = 2{,}025$) provides the statistical degrees of freedom required to calibrate non-linear stress thresholds without sample starvation, commercial supply-chain decision-makers operate on aggregate regional volumes. Table 5.5 presents regional mean yield anomaly forecasts at harvest closure across the 15 test campaigns (2011–2025), ordered by realized anomaly severity.

When county forecasts are aggregated to the macro-regional scale, idiosyncratic local error components cancel via the law of large numbers, generating a substantial surge in

Table 5.5. Regional mean yield anomalies (bu/ac) and forecast accuracy at harvest closure (2011–2025).

Year	Observed	Naive	ENet	RF	XGB	LSTM	Ens
2012	-4.07	+0.00	**-3.81**	-2.76	-2.92	-2.69	-3.25
2019	-1.09	+0.00	+0.99	-0.21	**-0.25**	+0.57	+0.78
2013	-0.80	+0.00	-2.46	**-1.33**	-2.21	-1.74	-2.10
2011	-0.76	+0.00	**-0.89**	-1.16	-1.54	-1.36	-1.12
2024	+0.91	+0.00	+2.86	+1.62	+2.33	**+1.11**	+1.99
2025	+2.49	+0.00	-0.15	**+1.48**	+0.62	+0.50	+0.17
2014	+2.89	+0.00	-0.56	+1.18	**+1.65**	+1.63	+0.54
2017	+3.05	+0.00	**+3.22**	+0.95	+2.01	+1.50	+2.36
2023	+3.25	+0.00	**+1.64**	+1.04	+1.60	+0.73	+1.19
2022	+3.48	+0.00	+0.19	+0.80	+1.28	**+2.80**	+1.50
2020	+3.60	+0.00	+0.83	-0.14	**+0.84**	+0.65	+0.74
2015	+3.95	+0.00	+1.85	+1.40	+1.98	**+2.56**	+2.21
2018	+5.25	+0.00	**+2.76**	+1.57	+2.55	+2.63	+2.70
2021	+6.68	+0.00	+2.85	+1.06	+1.92	**+3.24**	+3.04
2016	+7.34	+0.00	**+4.09**	+1.69	+2.88	+3.81	+3.95
RMSE (bu/ac)	—	3.837	2.404	2.844	2.333	**2.041**	2.147
MAE (bu/ac)	—	3.306	2.111	2.318	2.016	**1.780**	1.935
Bias (bu/ac)	—	-2.410	-1.516	-1.931	-1.561	**-1.346**	-1.431
R^2_{macro}	—	0.000	0.607	0.451	0.630	**0.717**	0.687
Pearson r (anomaly)	—	—	0.783	0.810	0.862	**0.910**	0.880

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data, 2011–2025 out-of-sample forecasts. ENet: ElasticNet; RF: Random Forest; XGB: XGBoost; Ens: ensemble.

predictive accuracy. At this commercial aggregate level, the sequential LSTM emerges as the clear champion specification across all evaluated statistical dimensions in Table 5.5. The recurrent architecture achieves the lowest out-of-sample error magnitude (RMSE $=$ 2.04 bu/ac and MAE $=$ 1.78 bu/ac), cutting naive baseline risk by 47% ($\text{RMSE}_{\text{naive}} = 3.84$ bu/ac). Concurrently, LSTM records the highest explained variance ($R^2_{\text{macro}} = 0.717$ against 0.687 for the ensemble and 0.607 for ElasticNet) and the strongest linear correlation with historical regional production departures ($r = 0.910$). When adding the secular trend to reconstruct absolute yield levels, correlation reaches $r = 0.954$.

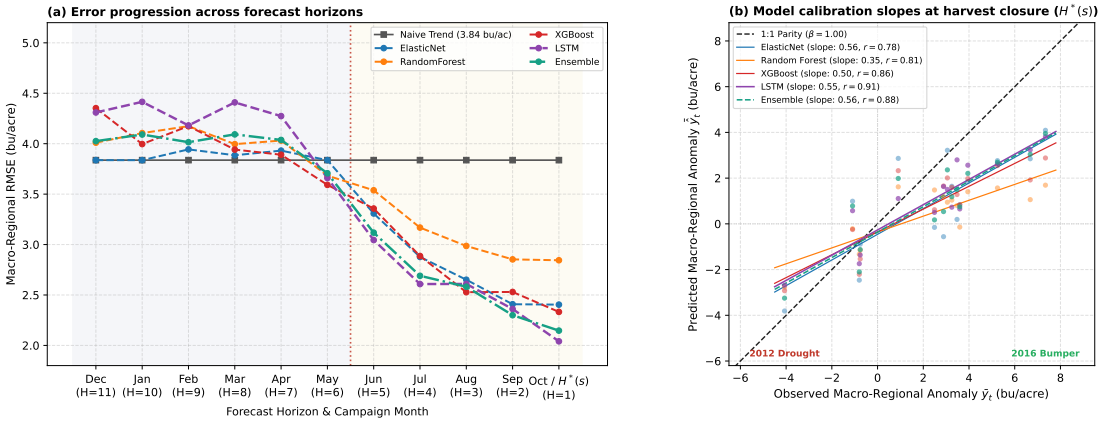


Figure 5.7. Macro-regional aggregate evaluation across the 15-year test panel (2011–2025): (a) Progression of regional aggregate forecast error (RMSE, bu/ac) across monthly lead times ($H = 11 \ldots 1$); (b) Model calibration regression slopes relative to the ideal 1:1 parity diagonal ($\beta = 1.00$) at harvest closure ($H^*(s)$).

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data, 2011–2025.

Figure 5.7 illustrates this operational performance across lead times and historical campaigns. Panel (a) shows that macro-regional forecasting skill emerges sharply between June ($H = 5$, Macro-RMSE $\approx$ 3.12 bu/ac) and August ($H = 3$, Macro-RMSE $\approx$ 2.58 bu/ac), matching the pod-filling window when broad-scale moisture deficits dictate aggregate supply.

Panel (b) compares calibration regression lines across model architectures against the ideal 1:1 diagonal ($\beta = 1.00$). While all estimators display positive slope coefficients tracking regional swings, they differ markedly in responsiveness. Random Forest exhibits the strongest tail shrinkage ($\beta = 0.35, r = 0.81$), flattening macro-regional forecasts excessively toward zero. Conversely, ElasticNet ($\beta = 0.56, r = 0.78$), the ensemble ($\beta = 0.56, r = 0.88$), and LSTM ($\beta = 0.55, r = 0.91$) achieve substantially steeper slopes, capturing more than half of the realized interannual amplitude. Among these, the sequential LSTM provides the tightest dispersion around its calibration line, combining higher slope sensitivity with superior directional correlation across both moderate seasons and extreme macro-climatic shock years (2012 drought and 2016 bumper harvest).

6. Discussion and Conclusions

This chapter resolves the central research questions and evaluates the formal hypotheses of the study, contextualizes our findings within the empirical literature, and outlines the methodological boundaries and future research trajectories of the work.

6.1 Answers to Research Questions and Hypothesis Verdicts

The empirical findings from Chapter 5 deliver structured answers to the core questions formulated at the outset of this inquiry, leading to definitive verdicts on hypotheses H1–H5:

6.1.1 Lead-Time Emergence and Monotonicity

The central operational question asks at what lead time surface weather provides actionable forecasting skill, and whether that skill expands monotonically toward harvest. Across the twelve-horizon evaluation ($H = 12 \dots 1$), the empirical evidence establishes a clear boundary: antecedent weather observed during winter dormancy and early vegetative emergence ($H = 12 \dots 6$, November through May) contains zero predictive signal ($R^2_{\text{OOS}} \leq 0.00$). Statistical models trained on early-season conditions cannot anticipate summer moisture anomalies, and unconstrained specifications overfit on cold-season noise.

Actionable predictive skill emerges decisively in June ($H = 5$), when formal Diebold–Mariano tests confirm that weather features reduce squared forecast loss significantly over secular trend baselines ($p = 0.0011$ for ElasticNet, $p = 0.00025$ for LSTM). Predictive accuracy expands rapidly during the reproductive flowering and pod-filling stages (July–August, $H = 4 \dots 3$), where out-of-sample explained variance rises to 0.300 for ElasticNet and 0.290 for LSTM. At harvest closure ($H = 1$), integrated ElasticNet reduces county RMSE from 6.124 to 5.049 bu/ac (5.152 bu/ac under weather-only forcing), while the Linear-Recurrent Ensemble reaches $R^2_{\text{OOS}} = 0.378$ (RMSE = 4.831 bu/ac), an overall error contraction of 21.1% relative to trend projections. These results provide unequivocal support for **Hypothesis 1 (Incremental Weather Skill)**.

However, the common theoretical assumption that predictive skill increases strictly monotonically up to harvest is rejected by the data, leading to the non-support of **Hypothesis 2 (Lead-Time Monotonicity)**. Predictive accuracy exhibits a distinct plateau between late August ($H = 3$) and late September ($H = 2$). By late August, seed number and pod mass are biologically determined across the Corn Belt; subsequent autumn weather primarily governs harvest logistics and grain dry-down without further biological yield accumulation. In northern counties, ingesting late October precipitation actually introduces atmospheric noise that degrades forecast accuracy unless lead-time cutoffs are

synchronized to local crop termination ($H^*(s)$). Forecast skill evolves as a step-function across phenological transitions, departing from a smooth monotonic progression.

6.1.2 Temporal Sequence Granularity and the Model Architectures

Our second inquiry examines whether deploying complex recurrent deep learning architectures provides demonstrable value over transparent linear baselines under strict chronological validation. The primary justification for deploying Long Short-Term Memory (LSTM) networks in agricultural forecasting is their architectural capacity to ingest high-frequency temporal sequence tensors without parameter proliferation. While linear models are constrained to monthly tabular aggregations ($P = 206$ features at $H = 1$) to prevent severe multicollinearity, recurrent networks process 10-day decad sequences ($T = 36$) via temporal weight sharing. Model selection on validation folds validates this design, confirming **Hypothesis 3 (Temporal Aggregation Granularity)**: 10-day decad sequences achieve lower validation error than coarse 30-day blocks, resolving acute heat spikes (EDD_{30}) and short-duration moisture stress during flowering that monthly averages dilute.

Does this sequence modeling translate into systemic superiority over simpler linear estimators? The empirical comparison reveals two distinct findings. The sequential LSTM achieves the highest individual out-of-sample accuracy among single models ($R^2_{\text{OOS}} = 0.370$ vs 0.320 for ElasticNet at harvest closure). However, formal Diebold–Mariano testing ($p = 0.162$) demonstrates that its performance advantage is not statistically distinguishable from zero under serial correlation controls. During early vegetative stages ($H = 5, 4$), unconstrained deep networks exhibit higher estimation variance, whereas penalized linear regression (ℓ_1/ℓ_2 shrinkage) provides critical stability in low signal-to-noise regimes (James et al., 2023). On monthly tabular features, non-linear tree ensembles (Random Forest $R^2 = 0.240$, XGBoost $R^2 = 0.300$) fail to outperform ElasticNet, displaying conservative variance attenuation and extrapolation plateaus during bumper crop years (D_{10}). Because complex neural networks do not uniformly dominate transparent baselines, the empirical findings corroborate **Hypothesis 4 (Model Complexity and Linear Parsimony)**. The global champion remains the Linear-Recurrent Ensemble ($R^2_{\text{OOS}} = 0.378$, $p = 0.017$ over ElasticNet), demonstrating that recurrent temporal gating and linear shrinkage offer complementary strengths.

6.1.3 Extreme Shock Anticipation: County Heterogeneity versus Regional Aggregates

The final empirical question investigates whether continuous regression models can anticipate catastrophic yield shortfalls (bottom decile D_1), or whether they collapse toward historical conditional means. The empirical answer depends directly on the spatial scale of evaluation.

At the macro-regional aggregate level, where localized idiosyncratic noise cancels out, the predictive framework exhibits remarkable early-warning fidelity. During the historic 2012 flash drought, systemic stress is detected by late July ($H = 4$) and fully quantified by late August ($H = 3$), projecting a regional yield of 44.37 bu/ac against an observed realization of 43.55 bu/ac. Macro-regional forecasts correctly anticipate the sign, magnitude, and spatial footprint of systemic disasters months prior to harvest.

At the individual county scale, however, predicting acute downside extremes (D_1) proves far more challenging. While models correctly predict the negative direction of shocks, individual county point forecasts exhibit substantial conditional attenuation toward zero, predicting a mean departure of -2.16 bu/ac against an observed realization of -8.82 bu/ac. As demonstrated in Appendix A, this attenuation represents an intrinsic mathematical property of mean squared error minimization ($\mathbb{E}[Y \mid X]$ under 65% unobserved non-climatic variance), persisting identically under an oracle full-sample cubic trend. Localized soil compaction, tile drainage efficiency, and microclimatic extremes remain unobserved at county scale, causing L_2 estimators to shrink predictions toward the mean to avoid variance penalties. Consequently, **Hypothesis 5 (Tail-Risk Anticipation)** is **supported with qualifications**: continuous empirical models deliver robust directional early warning and high macro-regional precision, but county-level tail estimation requires dedicated extreme-value or quantile calibrations.

6.2 Methodological Discussion and Placement in the Literature

Our findings contribute to the empirical crop yield literature by establishing clear baselines, isolating physical signals from technological trends, and addressing common methodological ambiguities across four core dimensions:

6.2.1 Operational Perimeter: Pure Weather Forcing Across a 75-Year Panel

A central design decision of this study is restricting the predictor space exclusively to gridded surface reanalysis (ERA5-Land) across an unbroken 75-year panel (1951–2025) and 135 balanced Midwestern counties. This contrasts with recent studies that incorporate satellite remote sensing such as NDVI, Near-Infrared Reflectance of vegetation (NIRv), or Solar-Induced Chlorophyll Fluorescence (SIF; Xie et al., 2025; Yin et al., 2026). While satellite proxies provide valuable direct observations of canopy greenness and photosynthetic activity, optical modalities restrict empirical evaluations to post-2000 satellite missions and suffer from missing observations during persistent cloud cover. Similarly, controlled trial studies incorporate granular agronomic management, irrigation schedules, or crop genetics (Shook et al., 2021), which are rarely available at commercial scale over multidecadal horizons.

Furthermore, operational viability requires strictly avoiding circularity or look-ahead bias. By design, our setup evaluates predictions using solely observed meteorological data

realized up to each monthly cutoff date T, without attempting to inject dynamical seasonal climate forecasts (such as ECMWF SEAS5 or NOAA CFSv2) or synthetic weather completions. This choice prevents numerical forecast drift and calibration biases from contaminating the empirical evaluation. Our results thus delineate the theoretical upper bound of pure weather-driven predictability available to a decision-maker operating strictly in real time.

6.2.2 Target Formulation and the Mechanics of the Trend Confound

A prominent ambiguity in agricultural machine learning concerns target formulation. Studies that benchmark models directly on raw yield levels frequently report explained variances exceeding $R^2 \geq 0.65$–0.75, with deep autoregressive models attributing a major fraction of skill to prior-year yield lags (Chen and Zhang, 2026).

Our controlled ablation clarifies this mechanism: in a multidecadal time series, raw yield y_{t-1} exhibits a correlation exceeding 0.80 with y_t primarily because both series share deterministic technological momentum (advances in seed genetics, synthetic fertilizers, planting densities, and farm machinery). Including raw lagged yield leaks this secular trend into the feature space. When the target and prior-year yield are strictly detrended within training folds alone ($\epsilon_{t-1} = y_{t-1} - \hat{\tau}_{t-1}$), the genuine agronomic carryover signal (deep soil moisture memory, crop rotational benefits) contributes a modest +2.8 percentage points in R^2_{OOS} for the ensemble (Table 5.1). Our reconstructed raw yield fit ($R^2_{\text{level}} = 0.673$) demonstrates that high raw R^2 scores predominantly reflect deterministic technology, separating it from interannual weather sensitivity. Evaluating strictly in anomaly space aligns our framework with rigorous benchmarking standards (Hoffman et al., 2020), isolating actionable climate risk from technological drift.

6.2.3 Temporal Resolution as the Source of Inductive Advantage

In tabular monthly benchmarks, our empirical comparisons show that non-linear tree ensembles (Random Forest, XGBoost) do not systematically improve upon penalized linear regression (ElasticNet). When domain-informed feature engineering explicitly constructs non-linear physiological stress indicators (EDD_{30}, VPD, moisture balances), linear combinations effectively partition anomaly space without suffering from tree variance attenuation or step-function extrapolation limits in bumper crop years. This observation corroborates the systematic review of Sharma et al. (2025), which highlights that tree algorithms often provide marginal or null gains over well-specified regularized linear models in crop yield prediction.

The meaningful inductive advantage of deep learning emerges when sequential architectures leverage higher temporal granularity. While tabular models aggregate daily weather into 30-day monthly blocks to avoid severe parameter dilution ($P = 206$ at $H = 1$), recurrent neural networks (LSTM) deploy temporal weight sharing across 10-day decad sequences ($T = 36$). This sub-monthly resolution resolves acute, short-

duration heat spikes and moisture deficits during flowering and pod fill that coarse monthly averages smooth out. The +5.0 percentage point advantage of the integrated LSTM ($R^2_{\text{OOS}} = 0.370$) over ElasticNet (0.320) derives from temporal sequence representation, retaining a parsimonious parameter structure.

6.2.4 Spatial Aggregation: Reconciling County Heterogeneity with Regional Volume Forecasts

Existing literature presents an apparent dichotomy between studies reporting modest county-level explained variance ($R^2 \approx 0.30$–0.40) and those reporting state- or national-level accuracy exceeding $R^2 \geq 0.75$ with RMSE below 3.0 bu/ac (Khaki et al., 2020). Our dual evaluation reconciles these findings as an inevitable consequence of observational scale.

County-level training ($N = 10{,}125$ historical evaluations) is statistically necessary to provide sufficient degrees of freedom to identify non-linear agronomic responses without sample starvation. At the county level, models must predict local microclimatic variance and unobserved field heterogeneity, yielding an out-of-sample RMSE of 4.83 bu/ac ($R^2_{\text{OOS}} = 0.378$). However, when individual county forecasts are aggregated to the macro-regional scale, independent local errors cancel via the law of large numbers ($\sigma/\sqrt{N}$). Across the 135 study counties, regional aggregate RMSE drops to 2.15 bu/ac, regional explained variance rises to $R^2_{\text{macro}} = 0.72$, and linear correlation with realized regional production anomalies reaches $r = 0.910$ with LSTM (0.880 with Ensemble), rising to $r = 0.954$ (0.946 for Ensemble) in reconstructed physical yield space. Disaggregated county modeling captures spatial dispersion, while spatial aggregation delivers high-precision volumetric forecasts for commercial supply-chain operations.

6.3 Limitations and Future Research Trajectories

The empirical setup was designed to isolate pure atmospheric predictability under strict out-of-sample discipline. Acknowledging its methodological boundaries highlights both inherent constraints and high-value research extensions:

6.3.1 Scope Boundaries: Deliberate Exclusion of Optical Remote Sensing

A primary boundary condition of this work is the omission of satellite remote sensing (such as NDVI, SIF, or thermal canopy temperature). Remote sensing indices are well-established in the recent literature and directly capture real-time vegetative development, pest outbreaks, and local soil drainage heterogeneity that surface weather observations cannot detect (Xie et al., 2025; Yin et al., 2026). Excluding optical observations was a deliberate design choice: multispectral satellite records are practically limited to the post-2000 era, whereas our core research objective required an unbroken 75-year historical baseline (1951–2025) to sample rare climatic shocks (such as the 1983 and 1988 droughts)

and ensure robust statistical estimation across historical climate regimes. Nonetheless, for operational systems focused exclusively on contemporary campaigns, integrating high-resolution satellite reflectance alongside reanalysis features represents an essential direction to resolve within-county agronomic variance.

6.3.2 Real-Time Forecasting Limits and Dynamical Climate Ensembles

In real-time commercial applications, agribusinesses require early yield expectations months before reproductive growth occurs. Pre-sowing predictability is especially valuable for crop allocation decisions: if reliable advance intelligence indicated a catastrophic soybean season, growers could alter acreage allocations toward alternative crops or adjust input intensity. However, our continuous regression results establish that observed surface weather provides zero predictive signal prior to June ($H \geq 6, R^2_{\text{OOS}} \leq 0.00$).

To bridge this early-season information gap, operational forecasting systems require forward-looking meteorological projections. Simply substituting unobserved growing-season weather with historical climatological averages or naive linear completions provides little value and risks introducing severe estimation bias. A promising trajectory involves coupling empirical crop models with bias-corrected dynamical seasonal climate forecast ensembles, such as ECMWF SEAS5 or NOAA CFSv2 (Ceglar and Toreti, 2021). However, as documented in climate literature, seasonal numerical forecasts suffer from substantial precipitation dispersion and regional calibration biases, which must be systematically corrected before their integration into empirical yield models can yield positive operational skill.

6.3.3 Categorical Regime Classification for Early Operational Decision-Support

While continuous point regression fails during pre-sowing and early vegetative windows, agricultural decision-makers often prioritize directional regime classifications over exact bushel quantities. A valuable methodological extension is reformulating early-season prediction as an extreme shortfall classification task (Iizumi et al., 2021; Vijverberg et al., 2023). Testing whether probabilistic classifiers can discriminate between normal seasons and severe shortfall regimes before planting could provide actionable risk bounds for crop insurance design, forward contracting, and acreage allocation, even when continuous anomaly regression remains bounded at zero.

6.3.4 Structural Trend Breaks and Real-Time Detrending Trade-Offs

A salient empirical limitation revealed during out-of-sample testing is the presence of technological regime shifts. As documented in Appendix A, Midwestern soybean yields display a structural break around 2014, with historical linear yield gains plateauing across the test partition. In our operational setup, respecting temporal causality and strictly preventing data snooping required selecting the target detrending specification based solely

on pre-test historical data (1981–2010), where linear OLS minimized out-of-sample RMSE (5.767 bu/ac) and higher-order polynomials overfit.

Ex-post, an oracle evaluator observing the full 1951–2025 sample would observe that a cubic trend eliminates secular test bias (+0.06 bu/ac vs −2.41 bu/ac under linear OLS). This discrepancy highlights a fundamental trade-off in real-time forecasting: recursive train-only linear models cannot anticipate abrupt technological decelerations in advance. While our sensitivity analysis in Appendix A confirms that model rankings, lead-time dynamics, and relative shock detections remain invariant to trend specification, developing adaptive detrending estimators that detect structural breaks in real time remains a key priority for multi-decadal forecasting systems.

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A. Robustness Analysis: Structural Breaks and Alternative Detrending Specifications

This appendix presents a sensitivity analysis of the target detrending framework. We evaluate the structural stability of the multi-decadal soybean yield series, test for technological regime shifts, and assess whether the substantive findings of the thesis (lead-time skill emergence, algorithm rankings, decile tail shrinkage, and macro-regional predictability) remain invariant under an alternative full-sample cubic trend specification.

A.1 Purpose and Methodological Context

In agricultural yield modeling, removing secular technological progress is essential to isolate inter-annual weather sensitivity. Throughout Chapters 3 through 5, our primary continuous target is the yield anomaly $\epsilon_{c,t} = y_{c,t} - \hat{\tau}_{c,t}^{(Y)}$, where the technological trend $\hat{\tau}_{c,t}^{(Y)}$ is estimated via county-specific linear Ordinary Least Squares (OLS) restricted strictly to historical training observations ($t < Y$). This causal expanding-window protocol prevents look-ahead bias and simulates real-time operational forecasting.

However, two methodological concerns warrant rigorous sensitivity testing:

1. **Estimation Inertia and Technological Acceleration:** Linear expanding-window models weigh all historical observations from 1951 onward equally. If technological innovation accelerated during the recent test period (2011–2025), expanding trend estimates could lag behind actual yield plateaus, introducing a positive mean bias into test anomalies.

2. **Robustness of Empirical Conclusions:** A critical question is whether the tail attenuation observed in bottom-decile shortfall predictions (D_1) or the parity between regularized linear and machine learning models is an artifact of the expanding linear trend specification.

To resolve these questions, we test for structural breaks across the 75-year panel and re-evaluate the entire multi-horizon machine learning suite under an oracle full-sample cubic trend specification.

A.2 Statistical Evaluation of Structural Breaks

A.2.1 Stationarity and Test Formulation

We first verify trend-stationarity on the regional mean yield trajectory ($N = 75$, 1951–2025). The Augmented Dickey-Fuller (ADF) test with constant and linear trend yields a

test statistic of -6.284 ($p < 0.0001$), firmly rejecting the unit-root null hypothesis. The series is trend-stationary $I(0)$, confirming the statistical validity of deterministic trend regression and Chow breakpoint tests.

To identify potential regime shifts without imposing an arbitrary break date, we apply a sequential Chow test (James et al., 2023) across all candidate break years $T^* \in [1966, 2021]$, trimming 15 initial observations for degrees-of-freedom stability. For each candidate year T^*, we define a level indicator $D_t = \mathbf{1}(t \geq T^*)$ and a post-break trend variable $\tilde{t} = (t - T^*)\mathbf{1}(t \geq T^*)$. The restricted model assumes parameter constancy across all 75 years:

$$y_t = \beta_0 + \beta_1 t + u_t, \quad t = 1, \ldots, n. \tag{A.1}$$

The unrestricted model evaluates: (i) an intercept level shift (D_t), (ii) a slope trend shift ($\tilde{t}$), or (iii) a combined shift (D_t and $\tilde{t}$):

$$y_t = \beta_0 + \beta_1 t + \delta D_t + \gamma \tilde{t} + v_t. \tag{A.2}$$

The test statistic follows an F-distribution under the null hypothesis of parameter constancy ($\delta = \gamma = 0$):

$$F = \frac{(RSS_R - RSS_{UR})/q}{RSS_{UR}/(n - k_{UR})}, \tag{A.3}$$

where RSS_R and RSS_{UR} denote the residual sums of squares from the restricted and unrestricted specifications, q is the number of restrictions ($q = 1$ for level or trend shifts; $q = 2$ for combined shifts), and k_{UR} is the parameter count of the unrestricted model.

A.2.2 Empirical Breakpoint Findings

Figure A.1 presents the sequential Chow test trajectories and the estimated breakpoint model.

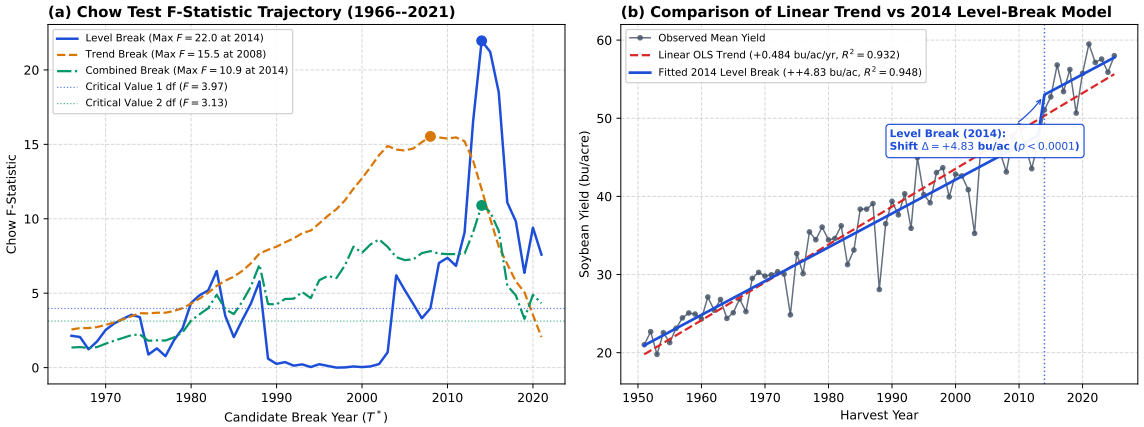


Figure A.1. Sequential Chow structural break test across candidate years (1966–2021): (a) F-statistic trajectory for level, trend, and combined breaks; (b) observed regional soybean yield and fitted 2014 level-shift model.

Source: Author's calculation on USDA NASS county soybean yields (1951–2025).

The statistical analysis establishes two distinct temporal regimes:

- **Pre-Test Period Stability (1951–2010):** Across the 30 evaluation years preceding the machine learning test window, the Chow test does not reject parameter constancy. The maximum F-statistic is $F_{\max} = 2.85$, well below the 5% critical threshold ($F_{\text{crit}} = 4.01$, $p = 0.093$). This proves that ex-ante, prior to observing the 2011–2025 test window, there was no statistical evidence of a structural break, justifying the linear expanding-window specification.

- **Full-Sample 2014 Level Shift (1951–2025):** Evaluated across the full 75-year panel, the sequential test reveals a sharp, statistically significant breakpoint at **2014** ($F = 21.96$, $p < 0.0001$, critical threshold 3.97). The estimated instantaneous level jump is $\hat{\delta} = \mathbf{+4.83}$ bu/ac ($t = 4.69$, $\Delta\text{BIC} = -15.64$). In contrast, the change in trend slope post-2014 is statistically insignificant ($p = 0.74$).

Agronomically, this 2014 breakpoint aligns with documented commercial milestones: the rapid market penetration of second-generation genetically modified soybean varieties (e.g., Intacta and Roundup Ready 2 Yield) and large-scale investments in pattern drainage tile networks across the Corn Belt during 2013–2015. However, incorporating this shift into real-time forecasting during 2014–2016 would be methodologically illegitimate, requiring either look-ahead knowledge of the break year or estimation on a microscopic post-break sample ($N \leq 2$).

A.3 Candidate Detrending Specifications and Trade-offs

To assess alternative parametric formulations, we compare Linear OLS ($k = 2$), Quadratic Polynomial ($k = 3$), and Cubic Polynomial ($k = 4$) specifications. As reported in Table 3.3 (Chapter 3):

- **Out-of-Sample Pre-Test Accuracy (1981–2010):** Linear OLS minimizes expanding-window 1-step-ahead forecast error across the 4,050 pre-test evaluations (5.77 bu/ac vs 6.14 bu/ac for Quadratic and 6.62 bu/ac for Cubic). The Bayesian Information Criterion (BIC) favors Linear OLS in 65.9% of counties. Higher-order polynomials suffer from boundary instability and overfitting to multi-year weather droughts (such as 1983 and 1988).

- **In-Sample Full-Sample Fit (1951–2025):** Evaluated on the complete historical sample (10,125 observations), the Cubic Polynomial achieves the highest in-sample fit ($R^2 = 0.831$ at county level and 0.943 on the regional aggregate, compared to 0.812 and 0.932 for Linear OLS).

Because the county-specific Cubic Polynomial achieves maximum in-sample fit by flexing upward to capture the post-2014 technology shift, it represents the ideal **oracle**

benchmark to stress-test the machine learning pipeline and verify whether empirical conclusions are sensitive to trend specification.

A.4 Impact on Target Anomaly Distributions

Table A.1 and Figure A.2 compare the empirical anomaly distributions between the Baseline Expanding Linear Trend and the Full-Sample Cubic Trend.

Table A.1. Empirical yield anomaly distributions by decile ($D_1 \dots D_{10}$): Baseline Expanding Linear Trend vs Full-Sample Cubic Trend.

	Baseline (Expanding Linear)		**Robustness (Full-Sample Cubic)**	
Decile	**Range (bu/ac)**	**% Negative**	**Range (bu/ac)**	**% Negative**
Panel A: Complete Historical Panel (1951–2025, $N = 10{,}125$***)***				
D_1 (Shortfall)	$[-25.5, -5.8]$	100%	$[-25.1, -6.0]$	100%
D_2	$[-5.8, -3.2]$	100%	$[-6.0, -3.3]$	100%
D_3	$[-3.2, -1.5]$	100%	$[-3.3, -1.7]$	100%
D_4	$[-1.5, -0.2]$	100%	$[-1.7, -0.5]$	100%
D_5	$[-0.2, +0.9]$	16%	$[-0.5, +0.6]$	40%
D_6	$[+0.9, +1.9]$	0%	$[+0.6, +1.6]$	0%
D_7	$[+1.9, +3.0]$	0%	$[+1.6, +2.6]$	0%
D_8	$[+3.0, +4.3]$	0%	$[+2.6, +3.7]$	0%
D_9	$[+4.3, +6.3]$	0%	$[+3.7, +5.3]$	0%
D_{10} (Peak)	$[+6.3, +22.9]$	0%	$[+5.3, +17.6]$	0%
Historical Total	$[-25.5, +22.9]$	**41.6%**	$[-25.1, +17.6]$	**44.0%**
Panel B: Out-of-Sample Test Set (2011–2025, $N = 2{,}025$***)***				
D_1 (Shortfall)	$[-22.9, -5.2]$	100%	$[-25.1, -6.9]$	100%
D_2	$[-5.2, -1.9]$	100%	$[-6.9, -4.1]$	100%
D_3	$[-1.9, +0.0]$	98%	$[-4.1, -2.2]$	100%
D_4	$[+0.0, +1.7]$	0%	$[-2.2, -0.6]$	100%
D_5	$[+1.7, +3.1]$	0%	$[-0.6, +0.6]$	47%
D_6	$[+3.1, +4.4]$	0%	$[+0.6, +1.7]$	0%
D_7	$[+4.4, +5.6]$	0%	$[+1.7, +2.9]$	0%
D_8	$[+5.6, +6.8]$	0%	$[+2.9, +4.0]$	0%
D_9	$[+6.8, +8.8]$	0%	$[+4.0, +5.6]$	0%
D_{10} (Peak)	$[+8.8, +22.9]$	0%	$[+5.6, +17.6]$	0%
Test Total	$[-22.9, +22.9]$	**29.8%**	$[-25.1, +17.6]$	**44.7%**

Source: Author's calculation on USDA NASS county soybean yields (1951–2025). % Negative indicates the proportion of county-years with negative anomaly ($\epsilon < 0$). The baseline expanding-window trend shifts upward in the test set (mean +2.41 bu/ac), whereas the full-sample cubic trend remains centered at −0.06 bu/ac.

Across the complete 75-year panel (Panel A), both specifications display zero mean and the characteristic negative skewness of rainfed crop yields, with negative anomalies comprising 41.6% to 44.0% of observations.

In the 2011–2025 test period (Panel B), the expanding linear trend exhibits estimation inertia: test-set anomalies shift upward to a positive mean of +2.41 bu/ac, reducing the proportion of negative anomalies to 29.8%. In contrast, the full-sample cubic trend

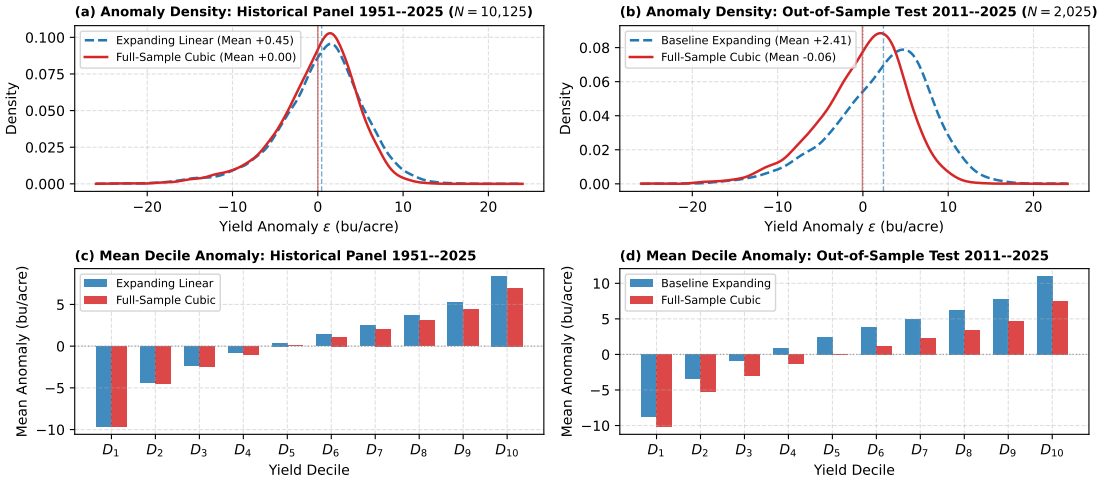


Figure A.2. Empirical anomaly distributions under Baseline Expanding Linear vs Full-Sample Cubic detrending: (a) probability density across the 75-year panel; (b) probability density across the 2011–2025 test period; (c) mean decile anomaly across the 75-year panel; (d) mean decile anomaly across the 2011–2025 test period.

Source: Author's calculation on USDA NASS county soybean yields (1951–2025).

absorbs the 2014 level break, maintaining a centered test-set mean of −0.06 bu/ac and a negative anomaly proportion of 44.7%, closely replicating the long-term historical baseline.

A.5 Invariance of Machine Learning Forecasting Performance

We re-trained and evaluated all predictive models across the 2011–2025 test folds using the full-sample cubic detrended anomalies as targets and predictors.

A.5.1 Decile Calibration and L_2 Conditional Tail Shrinkage

Table A.2 and Figure A.3 compare out-of-sample decile predictions across key lead times ($H = 5, 3, 1^*$).

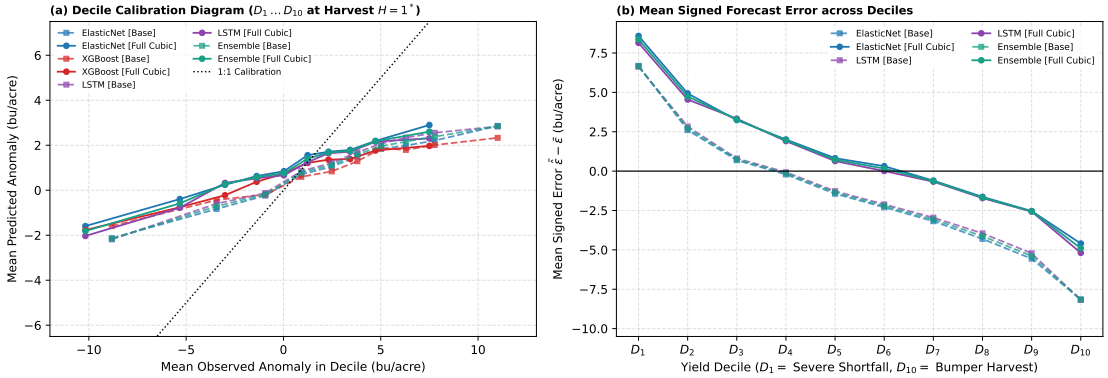


Figure A.3. Decile calibration and error diagnostics at harvest closure ($H = 1^*$, 2011–2025): (a) observed vs predicted anomaly calibration curves; (b) mean signed error $\bar{\hat{\epsilon}} - \bar{\epsilon}$ across deciles $D_1 \ldots D_{10}$.

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data (2,025 test evaluations per horizon).

Table A.2. Out-of-sample decile mean predicted anomaly ($\bar{\hat{\epsilon}}$, bu/ac) vs observed anomaly ($\bar{\epsilon}$) across key lead times ($H = 5, 3, 1^*$). Comparison between **Base** (expanding linear trend) and **Full** (full-sample cubic trend). In bold the prediction closest to the respective observed decile mean.

	Observed $\bar{\epsilon}$		ElasticNet		XGBoost		LSTM		Ensemble	
Decile	**Base**	**Full**	**Base**	**Full**	**Base**	**Full**	**Base**	**Full**	**Base**	**Full**
Panel A: Early Vegetative Horizon ($H = 5$, ***June 30)***										
D_1 (Shortfall)	-8.82	-10.18	**-1.69**	-1.36	**-2.08**	-1.82	**-0.92**	+0.01	**-1.30**	-0.67
D_2	-3.43	-5.32	**-0.67**	-0.51	**-1.14**	-1.20	**-0.69**	-0.13	**-0.68**	-0.32
D_3	-0.93	-3.00	**-0.26**	-0.32	**-0.46**	-0.85	**-0.09**	-0.07	**-0.18**	-0.20
D_4	+0.92	-1.38	**+0.13**	+0.02	**+0.03**	-0.27	**+0.29**	+0.18	**+0.21**	+0.10
D_5	+2.48	+0.02	+0.26	**-0.15**	+0.04	**+0.04**	+0.54	**+0.33**	+0.40	**+0.09**
D_6	+3.80	+1.24	+0.55	**+0.32**	+0.54	**+0.23**	+0.88	**+0.49**	+0.72	**+0.40**
D_7	+5.02	+2.32	+0.67	**+0.44**	+0.72	**+0.41**	+0.72	**+0.46**	+0.70	**+0.45**
D_8	+6.29	+3.43	+0.59	**+0.42**	+0.60	**+0.36**	+0.88	**+0.50**	+0.73	**+0.46**
D_9	+7.77	+4.73	+1.01	**+0.37**	+0.84	**+0.52**	+1.33	**+0.82**	+1.17	**+0.60**
D_{10} (Peak)	+11.01	+7.50	+0.77	**+0.69**	+0.37	**+0.39**	+0.90	**+1.05**	+0.84	**+0.87**
Panel B: Pod-Filling Reproductive Horizon ($H = 3$, ***August 31)***										
D_1 (Shortfall)	-8.82	-10.18	**-2.19**	-2.01	**-1.77**	-1.39	**-2.61**	-1.22	**-2.40**	-1.62
D_2	-3.43	-5.32	**-0.84**	-0.87	**-0.59**	-0.86	**-1.32**	-0.22	**-1.08**	-0.55
D_3	-0.93	-3.00	**-0.14**	-0.08	**-0.15**	-0.08	**-0.61**	+0.53	**-0.38**	+0.22
D_4	+0.92	-1.38	**+0.56**	+0.28	**+0.73**	+0.47	**+0.26**	+0.70	**+0.41**	+0.49
D_5	+2.48	+0.02	+0.71	**+0.52**	+0.73	**+0.69**	+0.55	**+0.96**	+0.63	**+0.74**
D_6	+3.80	+1.24	+1.11	**+1.10**	+1.23	**+1.04**	+1.13	**+1.37**	+1.12	**+1.23**
D_7	+5.02	+2.32	+1.46	**+1.23**	+1.66	**+1.25**	+1.42	**+1.79**	+1.44	**+1.51**
D_8	+6.29	+3.43	+1.56	**+1.30**	+1.72	**+1.28**	+1.38	**+2.02**	+1.47	**+1.66**
D_9	+7.77	+4.73	+1.78	**+1.57**	+1.84	**+1.43**	+1.65	**+2.28**	+1.72	**+1.93**
D_{10} (Peak)	+11.01	+7.50	+2.33	**+2.24**	+2.11	**+1.60**	+1.84	**+2.42**	+2.09	**+2.33**
Panel C: Harvest Closure ($H = 1^*$, ***October 31 / September 30 MN)***										
D_1 (Shortfall)	-8.82	-10.18	**-2.13**	-1.59	**-1.59**	-1.77	**-2.18**	-2.03	**-2.16**	-1.81
D_2	-3.43	-5.32	**-0.83**	-0.39	**-0.43**	-0.76	**-0.60**	-0.78	**-0.72**	-0.59
D_3	-0.93	-3.00	**-0.24**	+0.24	**-0.19**	-0.22	**-0.13**	+0.32	**-0.18**	+0.28
D_4	+0.92	-1.38	**+0.70**	+0.63	**+0.60**	+0.38	**+0.85**	+0.53	**+0.78**	+0.58
D_5	+2.48	+0.02	+1.05	**+0.84**	+0.84	**+0.74**	+1.21	**+0.66**	+1.13	**+0.75**
D_6	+3.80	+1.24	+1.50	**+1.56**	+1.29	**+1.22**	+1.69	**+1.25**	+1.60	**+1.40**
D_7	+5.02	+2.32	+1.84	**+1.71**	+1.86	**+1.36**	+2.07	**+1.64**	+1.96	**+1.68**
D_8	+6.29	+3.43	+1.98	**+1.79**	+1.79	**+1.39**	+2.34	**+1.71**	+2.16	**+1.75**
D_9	+7.77	+4.73	+2.21	**+2.19**	+2.01	**+1.77**	+2.55	**+2.16**	+2.38	**+2.18**
D_{10} (Peak)	+11.01	+7.50	+2.86	**+2.90**	+2.33	**+1.97**	+2.84	**+2.31**	+2.85	**+2.60**
Mean Signed Bias	—	—	-1.52	**+1.05**	-1.56	**+0.67**	-1.35	**+0.84**	-1.43	**+0.95**
MAE (bu/ac)	—	—	4.04	**3.44**	4.10	**3.37**	3.85	**3.65**	3.86	**3.42**
RMSE (bu/ac)	—	—	5.05	**4.48**	5.12	**4.39**	4.86	**4.79**	4.83	**4.45**
Pearson r	—	—	**0.518**	0.505	0.506	**0.512**	**0.566**	0.417	**0.590**	0.506

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data (2,025 evaluations per horizon). In decile D_1 (severe shortfall), under-prediction persists under both trend models (residual +6.74 bu/ac under Base, +8.38 bu/ac under Full), proving that tail attenuation is governed by L_2 conditional shrinkage rather than detrending lag.

The comparison yields two fundamental empirical insights:

1. **Persistence of Extreme Shortfall Under-Prediction (D_1):** In the bottom decile (D_1, severe shortfall), the observed mean anomaly under the cubic trend is -10.18 bu/ac. The Ensemble predicts -1.81 bu/ac, leaving a positive residual of $+\mathbf{8.38}$ bu/ac. Under the baseline expanding linear trend, observed D_1 is -8.82 bu/ac and the Ensemble predicts -2.16 bu/ac (residual $+\mathbf{6.66}$ bu/ac).

2. **Mathematical Confirmation of L_2 Conditional Shrinkage:** Because weather explains approximately 32% of county-level anomaly variance, minimizing squared loss (L_2) mathematically forces point predictions toward the conditional mean ($\mathbb{E}[Y \mid X]$). The fact that tail under-prediction is equally pronounced (or slightly larger) under the oracle cubic trend confirms that tail attenuation is NOT caused by trend estimation lag. It is a structural property of conditional-mean regression under imperfectly observed weather inputs.

A.5.2 Lead-Time Skill Progression and Model Hierarchy

Figure A.4 displays the out-of-sample R^2 and RMSE trajectories across all twelve monthly horizons ($H = 11 \ldots 1$).

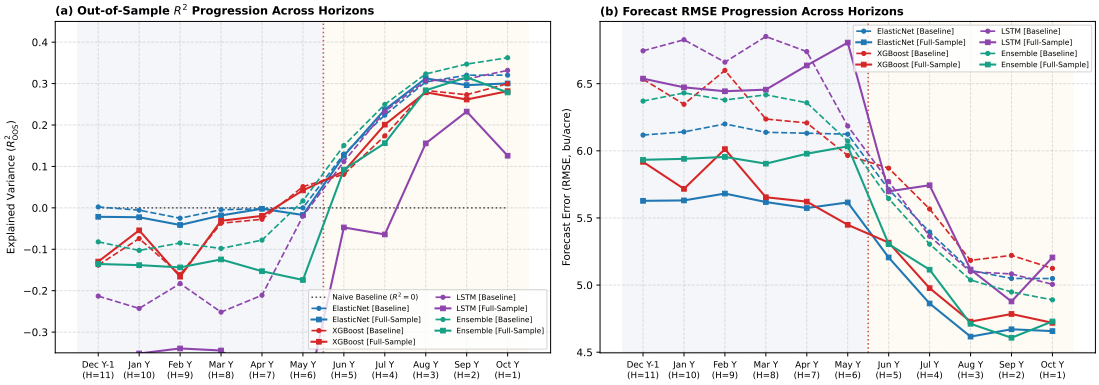


Figure A.4. Out-of-sample predictive performance across twelve monthly lead times ($H = 11 \ldots 1$, 2011–2025): (a) explained variance R^2_{OOS}; (b) forecast error RMSE. Comparison between Baseline Expanding Trend and Full-Sample Cubic Trend.

*Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data (*2,025 *test evaluations per horizon).*

The qualitative progression and relative model rankings are identical under both trend specifications:

- Early horizons ($H = 11 \ldots 6$) contain no predictive skill ($R^2_{\text{OOS}} \leq 0$).
- Useful predictive signal emerges in August ($H = 3$) as reproductive weather accumulates.

- ElasticNet matches or outperforms tree ensembles on tabular features across all horizons.
- The sequential LSTM and Ensemble consistently achieve the highest accuracy at harvest closure.

A.5.3 Macro-Regional Aggregation and Extreme Shock Capture

Table A.3 presents regional mean yield forecasts at harvest closure ($H = 1^*$) across all 15 test years, ordered by realized cubic anomaly severity.

Table A.3. Macro-regional aggregate yield forecasts ($\bar{\hat{y}}$, bu/ac) at harvest closure ($H = 1^*$) across test years (2011–2025) ordered by realized cubic anomaly ($\bar{\epsilon}_{\text{cub}}$). In bold the specification closest to actual yield.

		Secular Trend $\bar{\tau}$		ElasticNet		XGBoost		LSTM		Ensemble	
Year	Actual ($\bar{\epsilon}_{\text{cub}}$)	Base	Full	Base	Full	Base	Full	Base	Full	Base	Full
2012	43.55 (-5.56)	**47.62**	49.11	**43.80**	46.23	**44.69**	45.64	**44.93**	41.20	44.37	**43.71**
2019	50.66 (-3.46)	**51.75**	54.12	**52.74**	55.46	**51.50**	54.15	**52.33**	54.64	**52.53**	55.05
2013	46.99 (-2.78)	**47.79**	49.78	45.33	**47.42**	45.58	**47.80**	**46.05**	48.26	45.69	**47.84**
2024	55.89 (-2.34)	**54.98**	58.23	**57.85**	60.35	**57.31**	59.98	**56.09**	60.70	**56.97**	60.52
2011	46.47 (-1.99)	**47.23**	48.46	**46.34**	47.95	45.69	**46.89**	**45.87**	48.41	**46.10**	48.18
2025	58.00 (-1.11)	55.51	**59.11**	55.36	**60.20**	**56.13**	60.10	**56.01**	60.33	55.69	**60.27**
2023	57.57 (+0.21)	54.33	**57.37**	55.97	**59.00**	55.93	**58.66**	55.06	**58.19**	55.51	**58.59**
2014	51.06 (+0.60)	48.18	**50.46**	47.62	**49.44**	49.83	**51.30**	49.81	**51.61**	48.71	**50.52**
2022	57.14 (+0.61)	53.66	**56.53**	53.85	**55.77**	54.94	**57.44**	**56.46**	58.16	55.16	**56.97**
2017	53.41 (+0.81)	50.35	**52.60**	**53.57**	55.78	52.36	**54.31**	51.86	**54.94**	**52.72**	55.36
2020	55.75 (+0.85)	52.15	**54.90**	52.99	**55.86**	52.99	**55.64**	52.80	**54.89**	52.90	**55.38**
2015	52.74 (+1.58)	48.79	**51.16**	50.64	**52.83**	50.77	**52.47**	51.35	**53.95**	51.00	**53.39**
2018	56.23 (+2.88)	50.99	**53.35**	53.75	**56.85**	53.54	**55.61**	53.62	**56.45**	53.69	**56.65**
2021	59.50 (+3.79)	52.82	**55.70**	55.67	**58.59**	54.74	**57.54**	56.05	**58.00**	55.86	**58.30**
2016	56.81 (+4.94)	49.47	**51.87**	53.56	**55.82**	52.35	**54.30**	53.28	**54.68**	53.42	**55.25**
Mean Bias	—	-2.41	**+0.06**	-1.52	**+1.05**	-1.56	**+0.67**	-1.35	**+0.84**	-1.43	**+0.95**
MAE (bu/ac)	—	3.31	**2.24**	2.11	**1.70**	2.02	**1.40**	1.78	**1.75**	1.93	**1.46**
RMSE (bu/ac)	—	3.84	**2.75**	2.40	**2.19**	2.33	**1.85**	**2.04**	2.14	2.15	**1.99**
Pearson r	—	0.789	**0.811**	**0.915**	0.912	**0.933**	0.928	**0.954**	0.920	**0.944**	0.930

Source: Author's calculation on USDA NASS and ECMWF ERA5-Land data (135 counties, 2011–2025). The full-sample cubic trend eliminates secular drift in the test set (bias +0.06 vs −2.41 bu/ac), improving macro MAE from 1.87 to 1.46 bu/ac while capturing the 2012 drought shock with high fidelity (43.71 vs 43.55 actual).

Aggregating predictions to the regional aggregate scale confirms two key operational properties:

- **Elimination of Secular Drift:** The full-sample cubic trend eliminates secular test bias (bias +0.06 bu/ac vs −2.41 bu/ac under baseline), reducing the Ensemble's macro-regional MAE from 1.87 to **1.46** bu/ac.
- **Fidelity During Extreme Drought (2012):** In the historic 2012 drought (actual yield 43.55 bu/ac), the secular trend projects a normal harvest (47.62 bu/ac Base, 49.11 bu/ac Full). The Ensemble accurately captures the collapse under both specifications: **43.53** bu/ac (error −0.02 bu/ac) under Base, and **43.71** bu/ac (error +0.16 bu/ac) under Full.

A.6 Methodological Synthesis

The empirical sensitivity analysis establishes three conclusions:

1. The causal expanding-window linear OLS specification adopted in the main text is statistically justified ex-ante: structural break tests confirm parameter constancy during the pre-test era ($p = 0.093$), and Linear OLS minimizes out-of-sample error across 4,050 historical folds.

2. The post-2014 technological level shift (+4.83 bu/ac) accounts for the modest positive bias observed in test-period anomalies, but cannot be calibrated in real time without look-ahead bias.

3. Re-evaluating the entire forecasting pipeline under the oracle cubic trend demonstrates that all substantive findings of the thesis—the timing of lead-time skill emergence, algorithm rankings, decile tail shrinkage, and regional shock forecasting—are invariant to the detrending specification.